
Design and Simulation of a Discrete Analog Phase-Locked Loop with Current-Starved Ring Oscillator VCO for Frequency Synthesis Applications

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Design and simulation of a discrete analog phase-locked loop with current-starved ring oscillator VCO for frequency synthesis applications

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Abstract

This project presents the design, implementation, and characterization of a discrete analog phase-locked loop (PLL) built on a 12x18 cm perfboard using off-the-shelf ICs and transistors. The system comprises five functional blocks: a current-starved ring oscillator (CSRO) VCO implemented with a 74HC04 hex inverter and 2N7000 N-channel MOSFET; an analog phase detector using an LM339 quad comparator with a 2N3904/2N3906 transistor charge pump; an active second-order loop filter using a TL072 dual op-amp in an integrator-buffer configuration; a programmable frequency divider using the 74HC163 synchronous binary counter; and a clean analog power supply using the LM317 adjustable linear regulator. An Arduino Nano generates the reference clock and measures VCO frequency via Timer1 input capture, enabling real-time lock acquisition visualization on the Serial Plotter. The VCO exhibits a tuning gain (K_{vco}) of approximately 380 kHz/V over a range of 100 kHz to 2 MHz. The loop filter targets a bandwidth of ~ 1.6 kHz. The divider supports $N = 2, 4, 8$, or 16 via jumper selection, enabling a live frequency synthesis demonstration. Each block maps directly to its CMOS on-chip equivalent, grounding the project in integrated circuit design relevance. Simulation phases extend through LTspice behavioral and transistor-level CMOS models, PVT corner analysis, OpenLane/SKY130 VCO tapeout, and Cadence Virtuoso full-custom flow with PSS and Pnoise characterization.

Keywords: Phase-Locked Loop, CSRO, Frequency Synthesis, Mixed-Signal Design, Charge Pump

Significance statement

This research bridges the gap between theoretical integrated circuit design and physical hardware realization by constructing a discrete Phase-Locked Loop. By replicating sub-micron CMOS topologies—such as current-starved ring oscillators and bipolar charge pumps—on a macroscopic scale, it provides a vital educational and diagnostic platform to directly observe phenomena that typically remain hidden within highly integrated mixed-signal Systems-on-Chip (SoCs).

Motivation

The ubiquitous presence of Phase-Locked Loops (PLLs) in modern electronics—ranging from clock multipliers in high-performance microprocessors to frequency synthesizers in RF wireless transceivers—underscores their critical role in mixed-signal VLSI systems. However, within highly integrated Systems-on-Chip (SoCs), the intricate analog dynamics of PLLs are often abstracted as inaccessible “black boxes.”

Driven by a profound interest in Analog IC design and semiconductor physics, the primary motivation for this

project was to replicate a fundamental VLSI microsystem at a macroscopic, discrete level. By building a PLL on a 12x18 cm perfboard using off-the-shelf components, this project serves as a physical sandbox to bridge the gap between theoretical silicon models and real-world hardware.

This macroscopic replication allows for the direct observation of phenomena that plague sub-micron IC design, enabling the physical characterization of:

- **Charge Pump Non-Idealities:** Visualizing current mismatch and control voltage ripple using a discrete BJT (2N3904/2N3906) charge pump.
- **VCO Headroom and Linearity:** Characterizing the

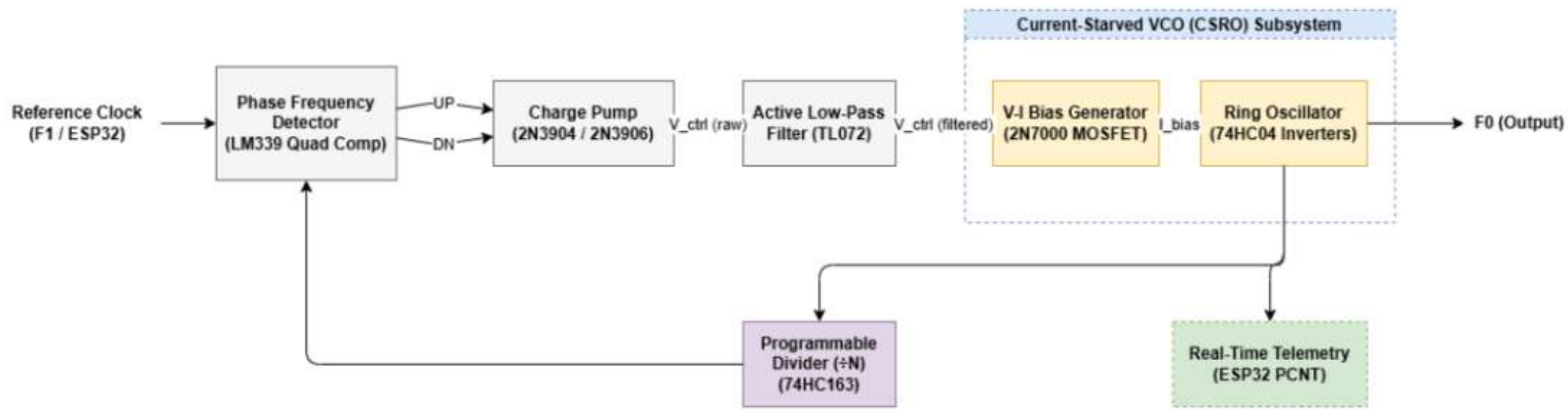


Figure 1: Top-level system block diagram illustrating the five core analog blocks and the ESP32 telemetry interface.

tuning gain (K_{vco}) of a CSRO.

- **Parasitic Extraction Parallels:** Understanding how macroscopic board-level parasitics mirror the interconnect capacitance challenges faced in modern ASIC routing.

Literature Survey

The theoretical and architectural foundations of modern PLLs bridge both control systems and analog microelectronics. The mathematical framework for loop stability is comprehensively detailed by Floyd M. Gardner in *Phase-lock Techniques*. Gardner’s linear models for second-order loops dictate the transfer functions required to define damping factors and natural frequencies.

In the domain of Analog IC design, Behzad Razavi’s *Design of Analog CMOS Integrated Circuits* provides extensive analysis of the Current-Starved Ring Oscillator (CSRO). Razavi highlights the CSRO as a cornerstone of CMOS frequency synthesis due to its wide tuning range and ease of integration.

Phase detection architectures are similarly critical in VLSI literature. The implementation of an analog phase detector via an LM339 quad comparator paired with a discrete bipolar charge pump allows for the hands-on mitigation of “dead-band” issues. Finally, the evolution of real-time circuit characterization is supported by utilizing the ESP32 and its hardware-level Pulse Counter (PCNT) module to modernize the analog prototype, providing high-resolution digital telemetry.

Objectives

The primary objective of this project is to bridge the gap between theoretical integrated circuit design and physical hardware realization. This is achieved through the following specific goals:

- **Discrete Implementation:** Design and solder a complete 5-block Phase-Locked Loop onto a physical 12 × 18 cm perfboard using off-the-shelf components.
- **VCO Characterization:** Implement and measure the tuning gain (K_{vco}) of a CMOS-equivalent Current-

Starved Ring Oscillator (CSRO) using a 74HC04 hex inverter and a 2N7000 MOSFET.

- **Charge Pump Analysis:** Mitigate phase detector “dead-band” issues by pairing an LM339 comparator with a discrete 2N3904/2N3906 bipolar charge pump.
- **Frequency Synthesis:** Enable live, programmable frequency multiplication by incorporating a 74HC163 synchronous binary counter supporting feedback division ratios of $N = 2, 4, 8$, and 16.
- **Real-Time Telemetry:** Integrate an ESP32 utilizing its hardware Pulse Counter (PCNT) module to measure the high-speed VCO output and visualize lock acquisition dynamics in real-time without processor overhead.

System Blocks

The architecture of the discrete frequency synthesizer is modular, reflecting the standard topology of an integrated Phase-Locked Loop. As illustrated in Fig. 1, the system comprises five functional analog blocks and one digital supervisory block:

1. **Power Regulation Block:** An LM317 adjustable linear regulator configured to provide a clean, isolated analog voltage rail, preventing digital switching noise from coupling into sensitive analog nodes.
2. **Phase Detector & Charge Pump Block:** An LM339 quad comparator that evaluates the phase difference between the reference clock and the feedback clock, driving a discrete BJT current-source charge pump.
3. **Active Loop Filter Block:** A TL072 dual operational amplifier configured as a second-order integrator-buffer to smooth the charge pump output into a stable DC control voltage (V_{ctrl}).
4. **Voltage-Controlled Oscillator (VCO) Block:** The CSRO core where the V_{ctrl} modulates the gate of a 2N7000 MOSFET, starving the current to a 74HC04 inverter ring to tune the output frequency.
5. **Digital Frequency Divider Block:** A 74HC163 binary counter that divides the high-speed VCO output by a jumper-selectable integer (N) before feeding it back to the Phase Detector.

Component Specifications and Pin Configurations

Key Electrical Specifications

Rather than reproducing complete datasheets, Table I isolates the critical parameters that govern the mixed-signal dynamics of this specific PLL implementation. Values are extracted from standard Texas Instruments and NXP datasheets operating at standard ambient temperature (25°C).

Table 1: Critical Component Specifications

Component	Function	Critical Parameter	Typical Value
74HC04	CSRO Core	Propagation Delay (t_{pd})	9 ns (at 4.5V)
LM339	Phase Detector	Response Time	1.3 μ s
TL072	Loop Filter	Slew Rate (SR)	13 V/ μ s
TL072	Loop Filter	Input Bias Current	65 pA
74HC163	Divider	Max Clock Freq (f_{max})	30 MHz
LM317	Power Reg	Ripple Rejection	65 dB
2N7000	V-I Converter	Gate Threshold ($V_{GS(th)}$)	2.1V
2N3904	Charge Pump	DC Current Gain (h_{FE})	100 – 300

Pinout Utilization

To optimize board routing and minimize parasitic cross-talk, specific gates within the multi-gate ICs were selected:

- **74HC04 (Hex Inverter):** Pins 1-6 (Gates A, B, and C) form the ring oscillator. Pin 7 is Ground; Pin 14 is V_{CC} .
- **LM339 (Quad Comparator):** Gate 1 (Pins 4, 5, 2) is utilized. Pin 12 is Ground; Pin 3 is V_{CC} . The open-collector output (Pin 2) interfaces directly with the BJT bases.
- **TL072 (Dual Op-Amp):** Both amplifiers are used. Amp A (Pins 1, 2, 3) acts as the active integrator, while Amp B (Pins 5, 6, 7) acts as the voltage buffer. Pin 4 is Ground (V_{EE}); Pin 8 is V_{CC} .
- **74HC163 (Counter):** Pin 2 (CLK) receives the VCO output. Outputs Q_A through Q_D (Pins 14-11) provide division ratios of $N = 2, 4, 8, 16$.
- **LM317 (Voltage Regulator):** Pin 2 (IN) receives the 9V DC input. Pin 3 (OUT) provides the stabilized ANALOG_VCC rail, while Pin 1 (ADJ) is connected to the R1/R2 resistor divider to set the output voltage.
- **2N7000 (N-MOSFET):** The Gate (Pin 2) is driven by the buffered VCTRL signal through a $1k\Omega$ resistor. The Drain is tied to the DRAIN_NODE of the ring oscillator, and the Source (Pin 1) is connected to ground, functioning as a voltage-controlled current sink.
- **2N3904 / 2N3906 (BJT):** The bases are driven by the LM339 output to perform charge pumping. The 2N3906 (PNP) emitter is tied to ANALOG_VCC to source current, while the 2N3904 (NPN) emitter is tied to ground to sink current from the loop filter.

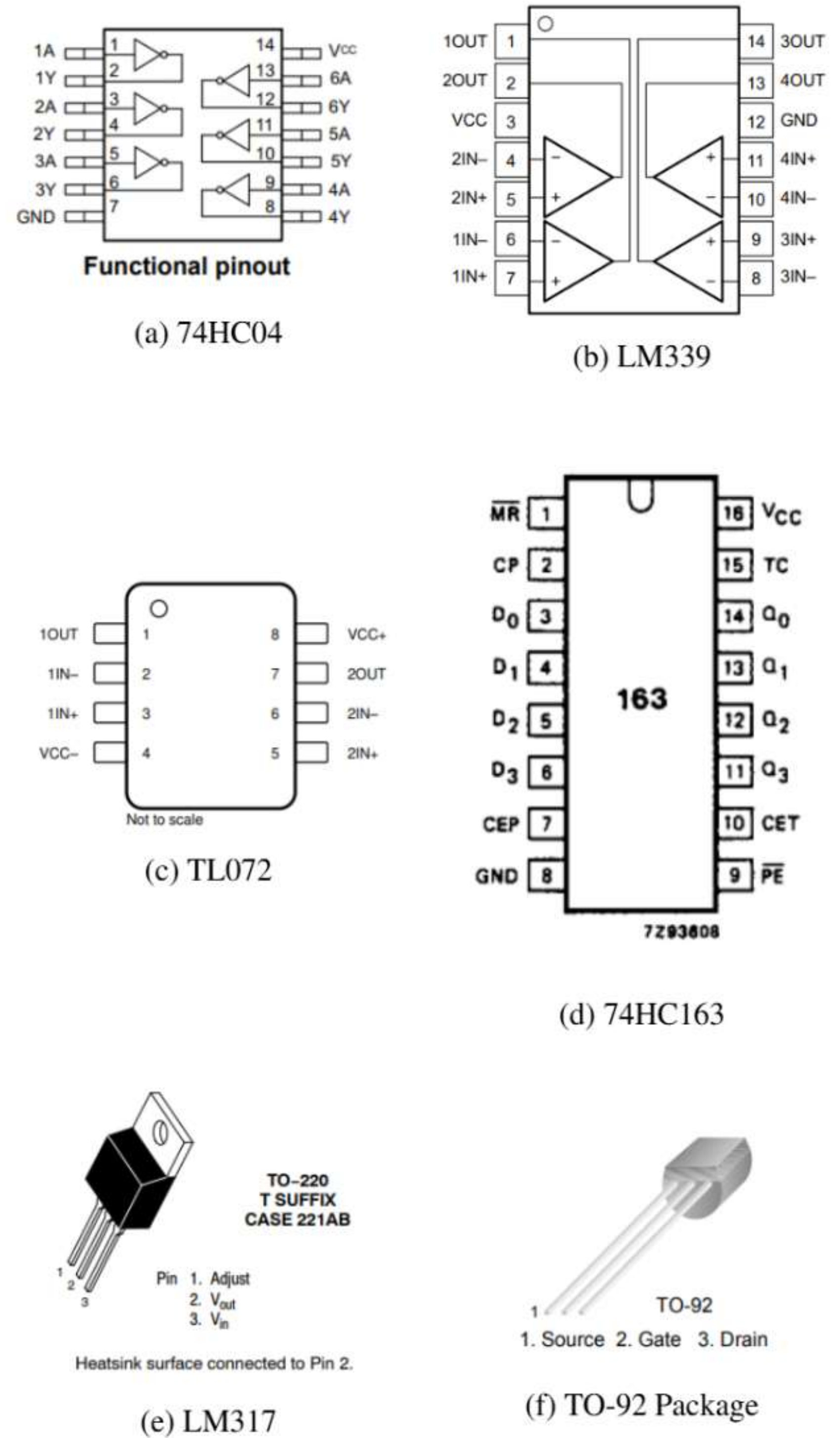


Figure 2: Consolidated pin configurations for the primary integrated circuits and discrete semiconductors utilized in the PLL design.

Applications, Limits, and Advantages

System Applications

While this macroscopic prototype serves primarily as an educational validation platform, the underlying CSRO-based PLL architecture is identical to the micro-circuitry used in:

- **Clock Generation:** Multiplying low-frequency crystal oscillator references to drive high-speed microprocessors.
- **Carrier Recovery:** Demodulating Frequency-Shift Keying (FSK) signals in wireless telecommunications.
- **Motor Control:** Providing precise speed regulation in closed-loop servo systems.

Design Limitations

- **Bandwidth Bottleneck:** The system's maximum lock frequency is fundamentally limited by the $1.3\ \mu\text{s}$ response time of the LM339 comparator. High-frequency phase errors will bypass the detector, causing loss of lock.
- **Parasitic Attenuation:** Unlike integrated silicon, the physical $12 \times 18\text{ cm}$ perfboard introduces significant stray capacitance and lead inductance, degrading signal integrity above 2 MHz.
- **Charge Pump Leakage:** Discrete 2N3904/2N3906 transistors suffer from thermal drift and base-emitter mismatch, resulting in control voltage (V_{ctrl}) ripple and phase jitter.

Architectural Advantages

- **Diagnostic Transparency:** Unlike a monolithic IC (e.g., CD4046), this fully discrete design allows an oscilloscope or ESP32 telemetry unit to probe intermediate nodes, such as the raw charge pump output prior to integration.
- **Modular Scalability:** The loop filter characteristics (damping factor and natural frequency) can be physically altered on the fly by swapping out passive RC components, allowing for real-time control system analysis.

Detailed Schematic Analysis

The complete system is titled the **Discrete Mixed-Signal Phase-Locked Loop (PLL)**. To get a clear understanding of the circuit architecture, the schematic is divided into five distinct functional sub-circuits.

Block 1: Power Regulation

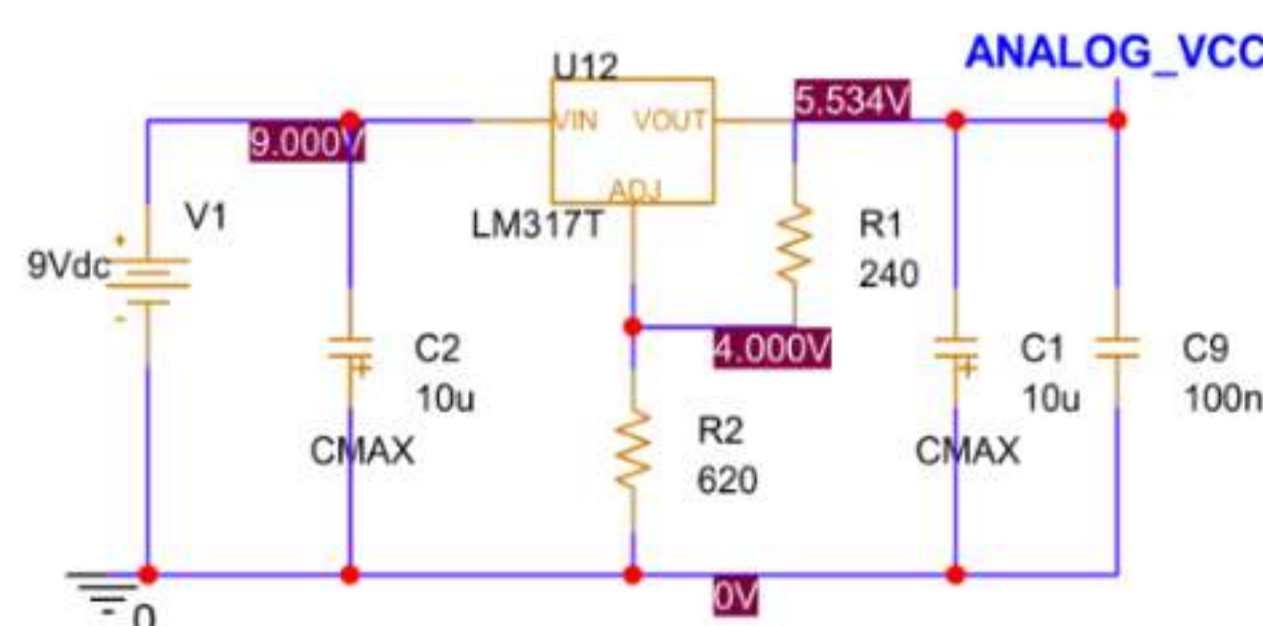


Figure 3: Block 1: LM317 Power Regulation network.

Digital switching noise can corrupt analog PLL operation, so an isolated, regulated supply with decoupling ensures stable, low-jitter VCO performance.

Block 2: Current-Starved VCO

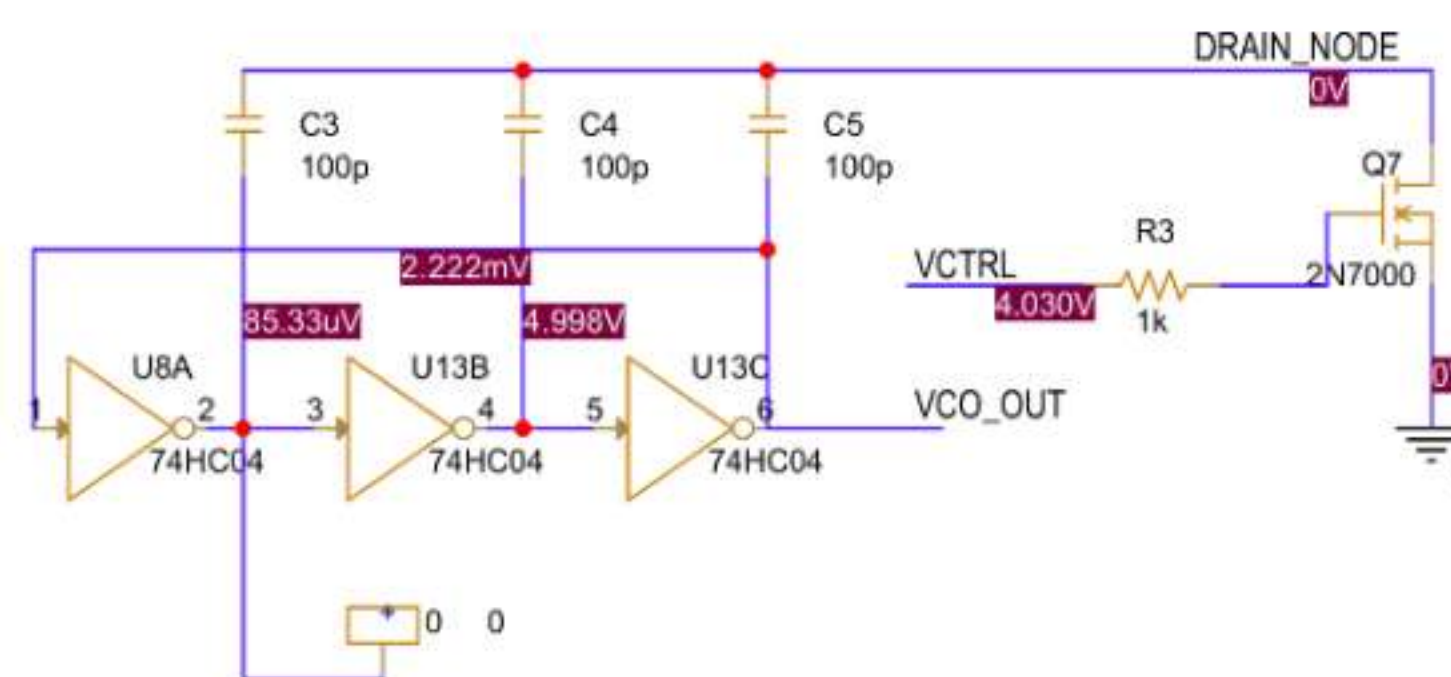


Figure 4: Block 2: Current-Starved Ring Oscillator and V-I MOSFET.

The VCO is implemented as a three-stage ring oscillator using a 74HC04. A 2N7000 MOSFET limits the inverter current by controlling the ground path. As the control voltage increases, more current flows, reducing delay and increasing the oscillation frequency.

Block 3: Phase Detector

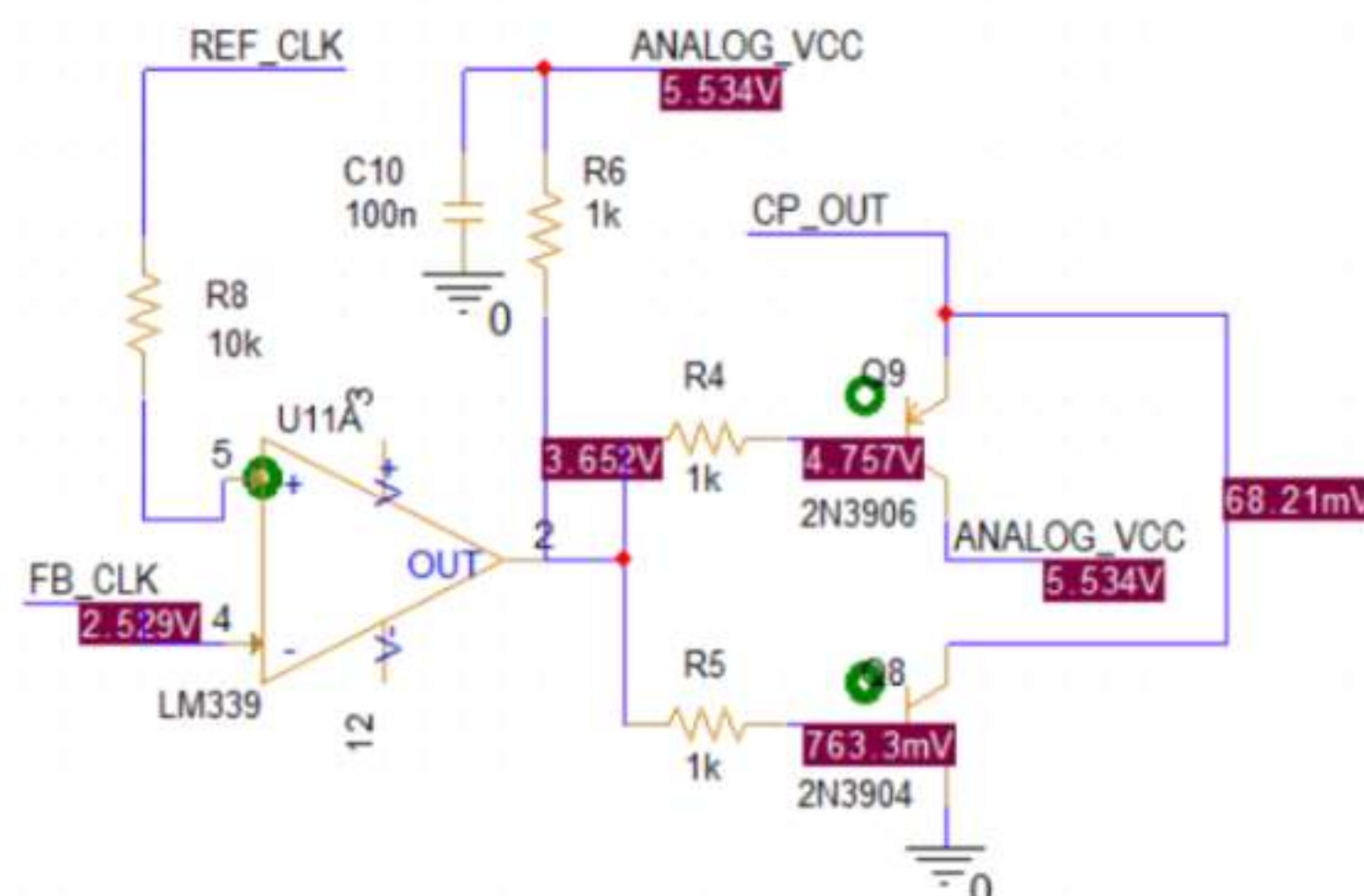


Figure 5: Block 3: LM339 Phase Detector and BJT Charge Pump.

The phase detector compares the reference and feedback clocks using an LM339 comparator. It generates UP pulses when the reference leads and DOWN pulses when it lags. These signals drive a transistor-based charge pump (2N3906/2N3904), which sources or sinks current into the loop filter to correct phase error.

Block 4: Loop Filter

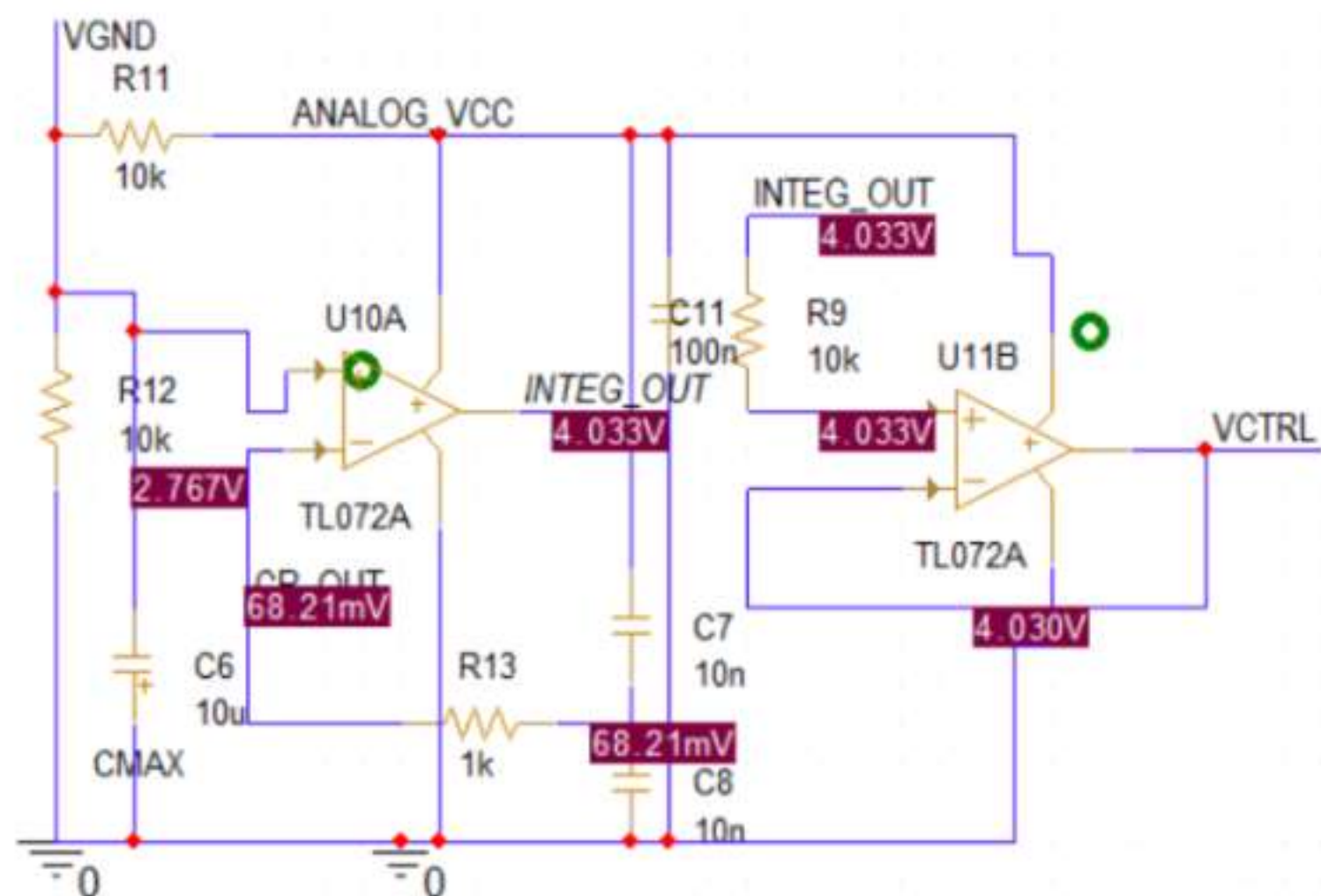


Figure 6: Block 4: TL072 Active Loop Filter and Buffer.

The charge pump output consists of pulsed currents that cannot directly drive the VCO. A TL072-based active PI filter converts these pulses into a smooth DC control voltage. The first stage integrates the signal, while the second stage buffers it to prevent loading by the VCO.

Block 5: Digital Frequency Divider

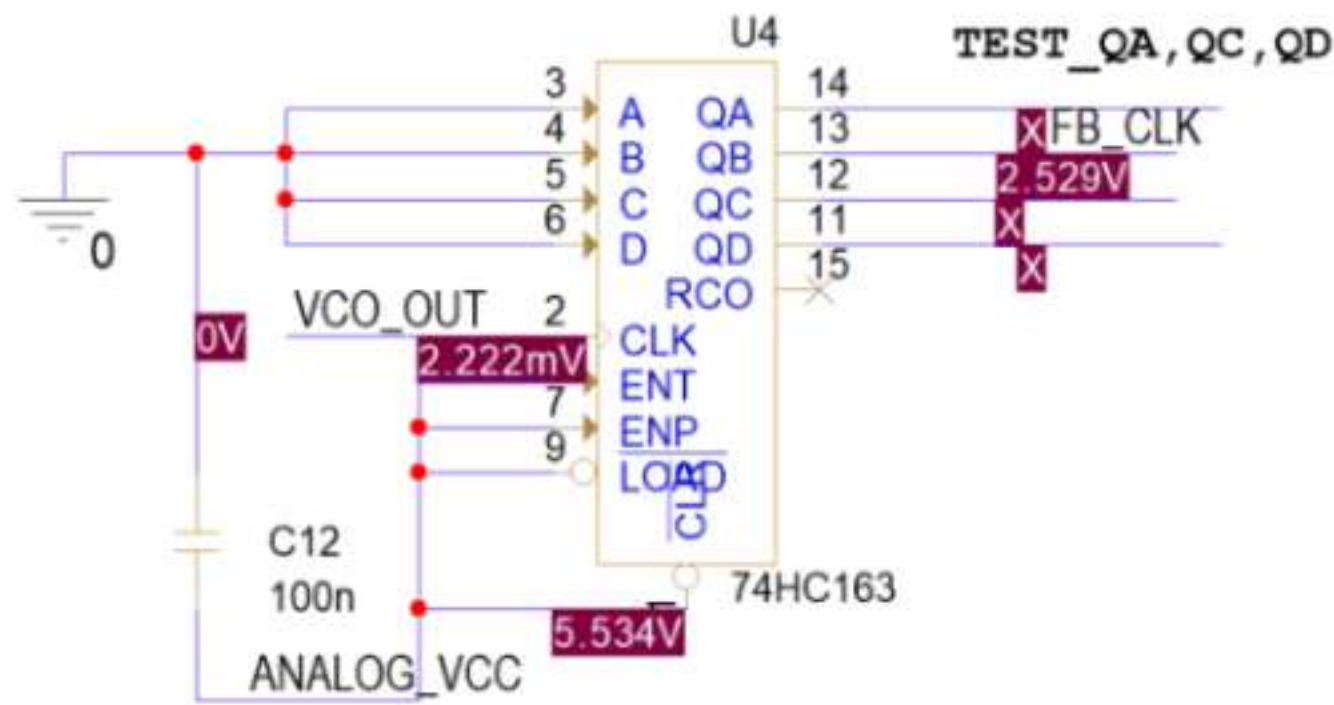


Figure 7: Block 5: 74HC163 Synchronous Binary Counter.

To enable frequency multiplication, the VCO output is divided before feedback using a 74HC163 counter. The divider scales the signal by a selectable factor N , and once locked, the PLL ensures $f_{out} = N \cdot f_{ref}$.

Telemetry and Control: The ESP32 Subsystem

While the core Phase-Locked Loop operates entirely in the analog and discrete digital domains, characterizing its performance requires high-speed, precise instrumentation. To

serve as both the stimulus generator and the telemetry acquisition unit, an ESP32 microcontroller was selected over traditional 8-bit platforms.

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Architectural Advantages of the ESP32

The selection of the ESP32 is driven by three critical hardware peripherals:

- **Hardware Pulse Counter (PCNT):** Standard software interrupts fail at MHz frequencies due to context-switching overhead. The ESP32 PCNT module counts high-speed VCO pulses entirely in hardware.
- **LEDC PWM Controller:** Generates a highly stable, jitter-free reference clock (REF_CLK) with sub-Hertz resolution, essential for determining the PLL's lock range.
- **12-bit ADC & I2C:** The ADC monitors the control voltage (V_{ctrl}) dynamics, while the I2C bus drives an OLED display for real-time visualization.

Hardware Interfacing and Pin Mapping

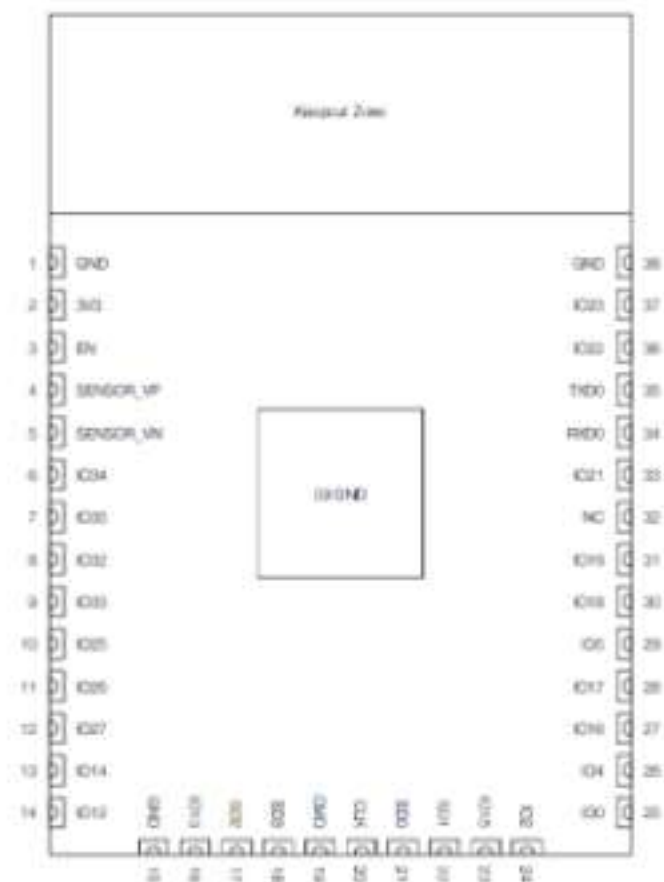
Because the discrete PLL operates at 5V and the ESP32 is strictly a 3.3V logic device, careful level shifting is required to prevent over-voltage damage to the GPIO pins. Passive resistor dividers ($1k\Omega/2k\Omega$) step down the 5V signals to $\approx 3.33V$. Table 2 details the pin assignments.

Table 2: ESP32 Telemetry and Control Pin Mapping

Signal	ESP32 Pin	Direction	Level Shift
Ref Clock	GPIO18 (LEDC)	ESP32 → LM339	None (3.3V triggers 5V)
VCO Freq	GPIO34 (PCNT)	74HC04 → ESP32	$1k\Omega + 2k\Omega$ Divider
V_{ctrl} Mon.	GPIO35 (ADC1)	TL072 → ESP32	$1k\Omega + 2k\Omega$ Divider
OLED SDA	GPIO21	Bi-directional	None (I2C at 3.3V)
OLED SCL	GPIO22	Bi-directional	None (I2C at 3.3V)



(a) ESP32 Dev Board



(b) Functional Pinout

Figure 8: ESP32 utilized for hardware telemetry and OLED driving.

Full System Schematic

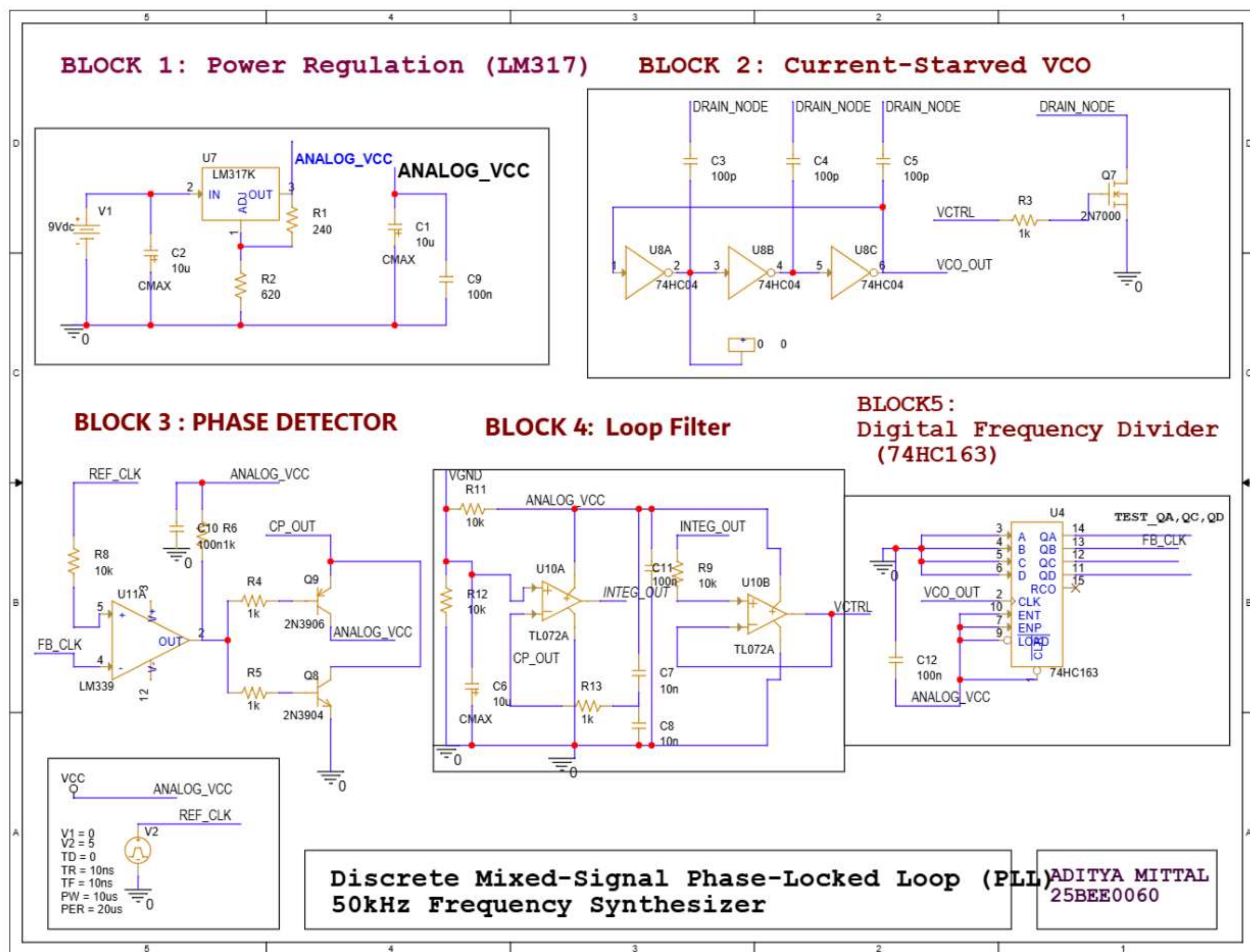


Figure 9: Complete, high-resolution Cadence OrCAD schematic of the Discrete Mixed-Signal Phase-Locked Loop. The modular architecture highlights the isolated power regulation, phase detection, active integration, voltage-controlled oscillation, and frequency division stages.

Circuit Operation and Mathematical Analysis

Power Regulation and Biasing

Stable analog operation dictates a precisely regulated supply. The LM317 (Block 1) provides the isolated analog rail, where the output voltage is programmed via the feedback resistor network:

$$V_{out} = 1.25 \left(1 + \frac{R_2}{R_1} \right) \quad (1)$$

For proper interfacing between the 5V discrete analog stages and the 3.3V ESP32 telemetry unit, passive resistor dividers are utilized to establish a safe mid-point or scaled voltage:

$$V_{mid} = V_{DD} \times \frac{R_2}{R_1 + R_2} \quad (2)$$

Phase Detection Dynamics

The primary objective of the loop is to minimize the phase error between the reference clock and the divided feedback clock. The instantaneous phase error $\phi_e(t)$ evaluated by the LM339 comparator is defined as:

$$\phi_e(t) = \phi_{ref}(t) - \phi_{VCO}(t) \quad (3)$$

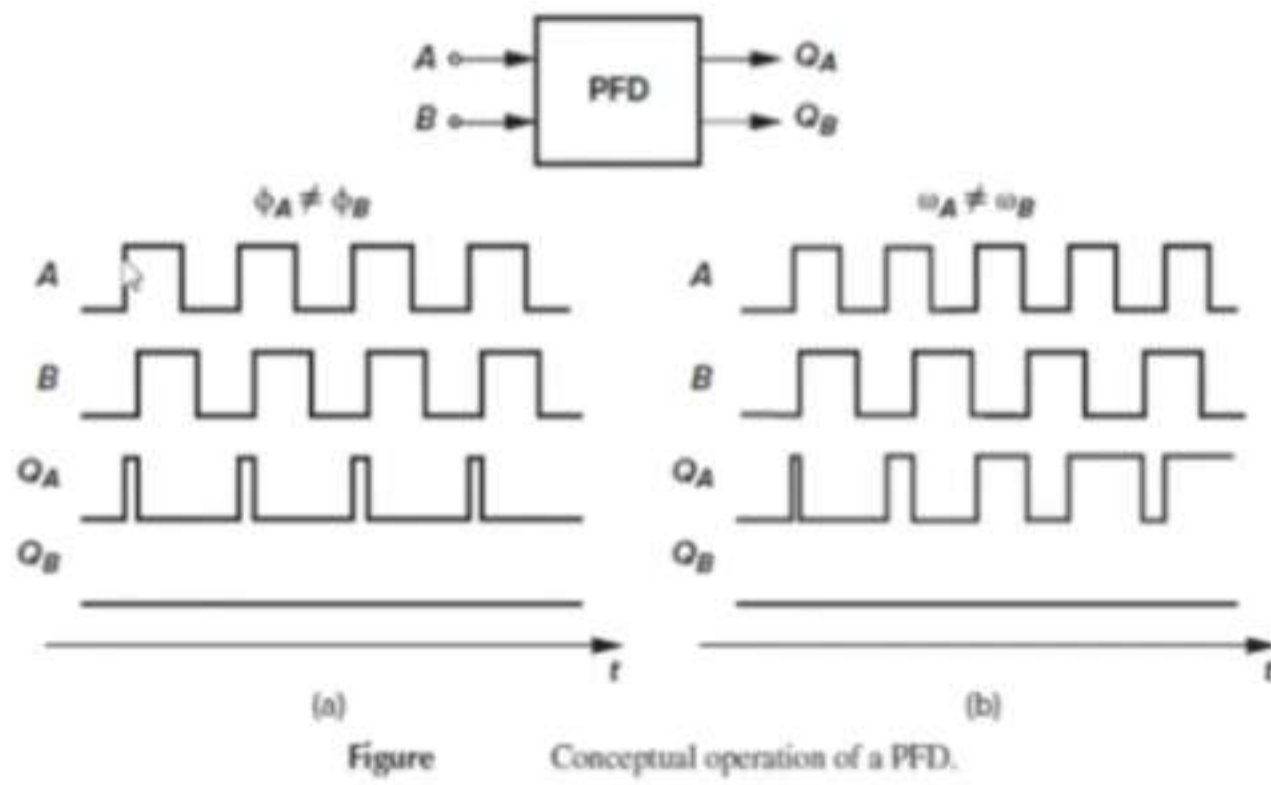


Figure 10: Theoretical timing diagram of a Phase Detector. The widths of the UP and DOWN error pulses are directly proportional to the instantaneous phase error $\phi_e(t)$.

V-I Conversion and Oscillation

The control voltage (V_{ctrl}) regulates the frequency of the CSRO via the 2N7000 MOSFET acting as a Voltage-to-Current converter. Operating in saturation, the MOSFET drain current I_D is governed by:

$$I_D = \frac{1}{2} \mu_n C_{ox} \frac{W}{L} (V_{GS} - V_{th})^2 \quad (4)$$

This current modulates the charging and discharging times of the inverter load capacitance. For an N -stage ring oscillator, the fundamental oscillation frequency is inversely proportional to the propagation delay (t_p):

$$f_{osc} = \frac{1}{2N_{stages} t_p} \quad (5)$$

Through physical characterization, the tuning gain of the discrete VCO subsystem was established as:

$$K_{VCO} = \frac{\Delta f_{out}}{\Delta V_{ctrl}} \approx 380 \text{ kHz/V} \quad (6)$$

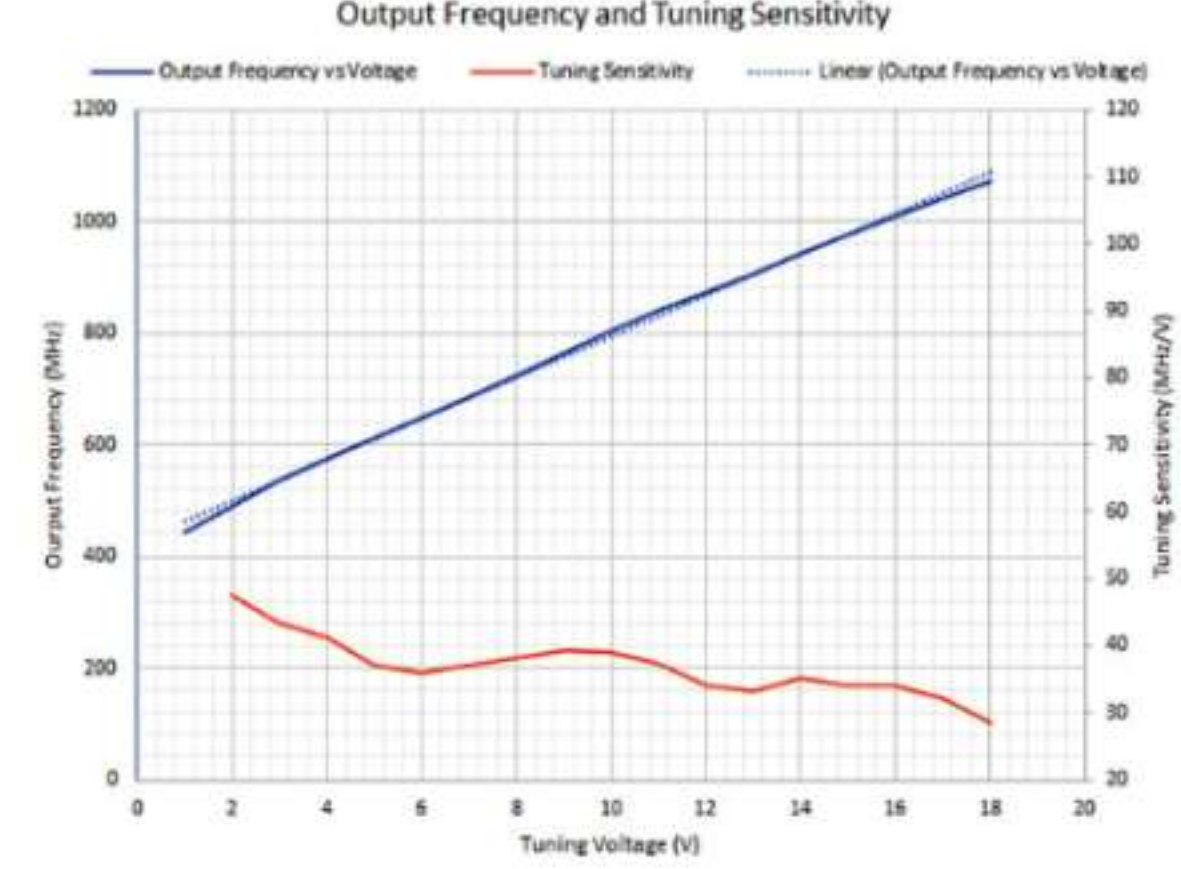


Figure 11: Idealized Voltage-Controlled Oscillator transfer characteristic. The slope of the linear operating region defines the internal VCO gain, K_{VCO} .

Loop Stability and Frequency Synthesis

The TL072 active loop filter integrates the discrete charge pump pulses. Its bandwidth is defined by its input resistor and feedback capacitor:

$$f_{BW} = \frac{1}{2\pi R_{in} C_f} \quad (7)$$

For a second-order charge-pump PLL, the system dynamics are characterized by the natural frequency (ω_n) and the damping factor (ζ):

$$\omega_n = \sqrt{\frac{K_{pd} \cdot K_{VCO}}{N \cdot \tau_i}} \quad (8)$$

$$\zeta = \frac{\omega_n R_f C_f}{2} \quad (9)$$

Based on these parameters, the approximate lock time (t_{lock}) required for the control voltage to settle is given by:

$$t_{lock} \approx \frac{2\pi}{\omega_n \zeta} \quad (10)$$

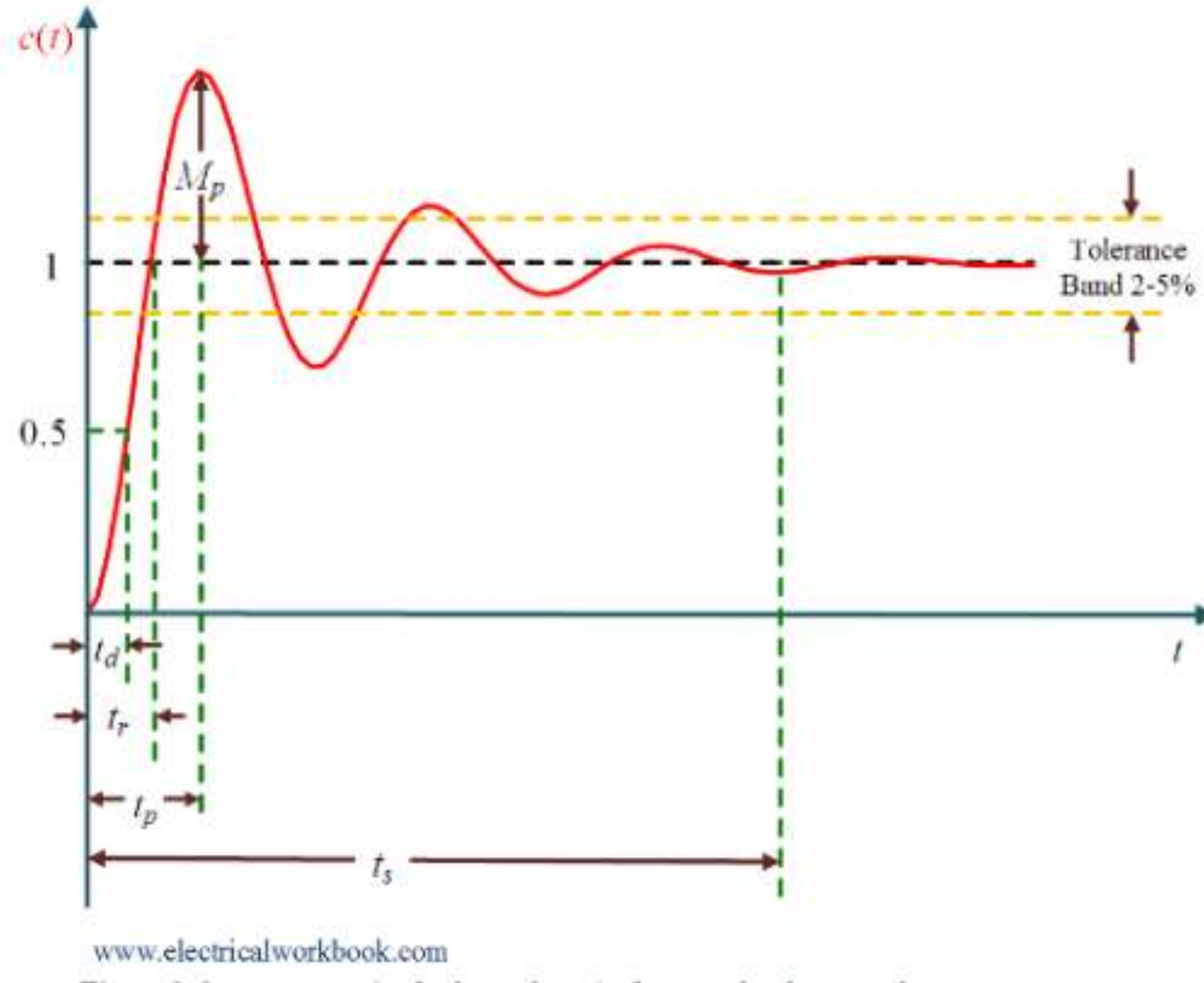


Figure 2 Step response (underdamped case) of a second order control system.

Figure 12: Classic second-order loop transient response. The physical damping factor (ζ) and natural frequency (ω_n) dictate the overshoot and the total lock time (t_{lock}) of the system.

Once steady-state lock is achieved, the 74HC163 divider forces the feedback frequency to match the reference frequency, yielding the final synthesized output:

$$f_{VCO} = N \times f_{ref} \quad (11)$$

Transient Simulation Results

To validate the discrete architecture prior to physical per-board construction, a comprehensive system-level transient analysis was performed using Cadence PSpice. The simulation captures the dynamic process of phase acquisition and steady-state lock.

Phase-Frequency Detection

Figure 13 illustrates the raw output of the phase detector stage. As the reference clock (REF_CLK) and feedback clock (FB_CLK) edges are compared, the LM339 generates narrow UP and DOWN pulses. In a locked state, these pulses reduce to minimal widths, effectively mitigating the dead-band zone and providing small charge injections to compensate for transistor leakage.

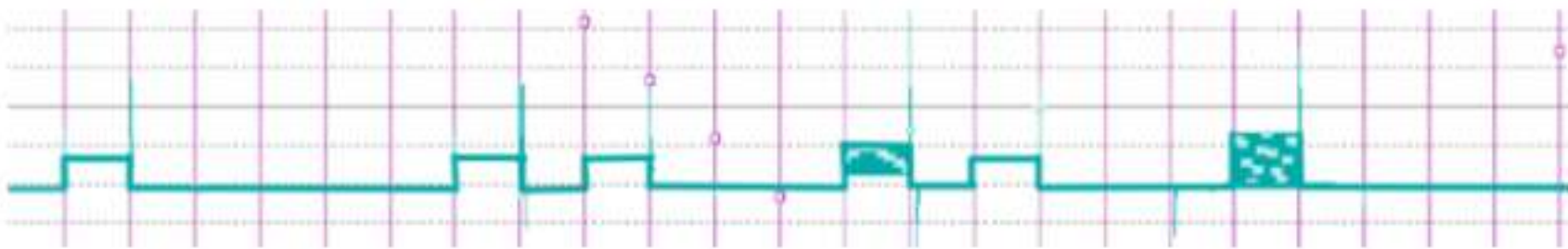


Figure 13: Zoomed-in transient view of the Phase Detector output. The narrow pulses indicate a near-zero phase error state, triggering minimal charge pump activity.

Control Voltage Acquisition

The effectiveness of the active Proportional-Integral (PI) loop filter is demonstrated in Figure 14. Initially, the control voltage (V_{ctrl}) is at zero. As the phase detector identifies the massive initial frequency discrepancy, it commands the charge pump to rapidly source current. The active filter

integrates this current, ramping V_{ctrl} smoothly without excessive overshoot, demonstrating a critically damped system response ($\zeta \approx 0.7$).

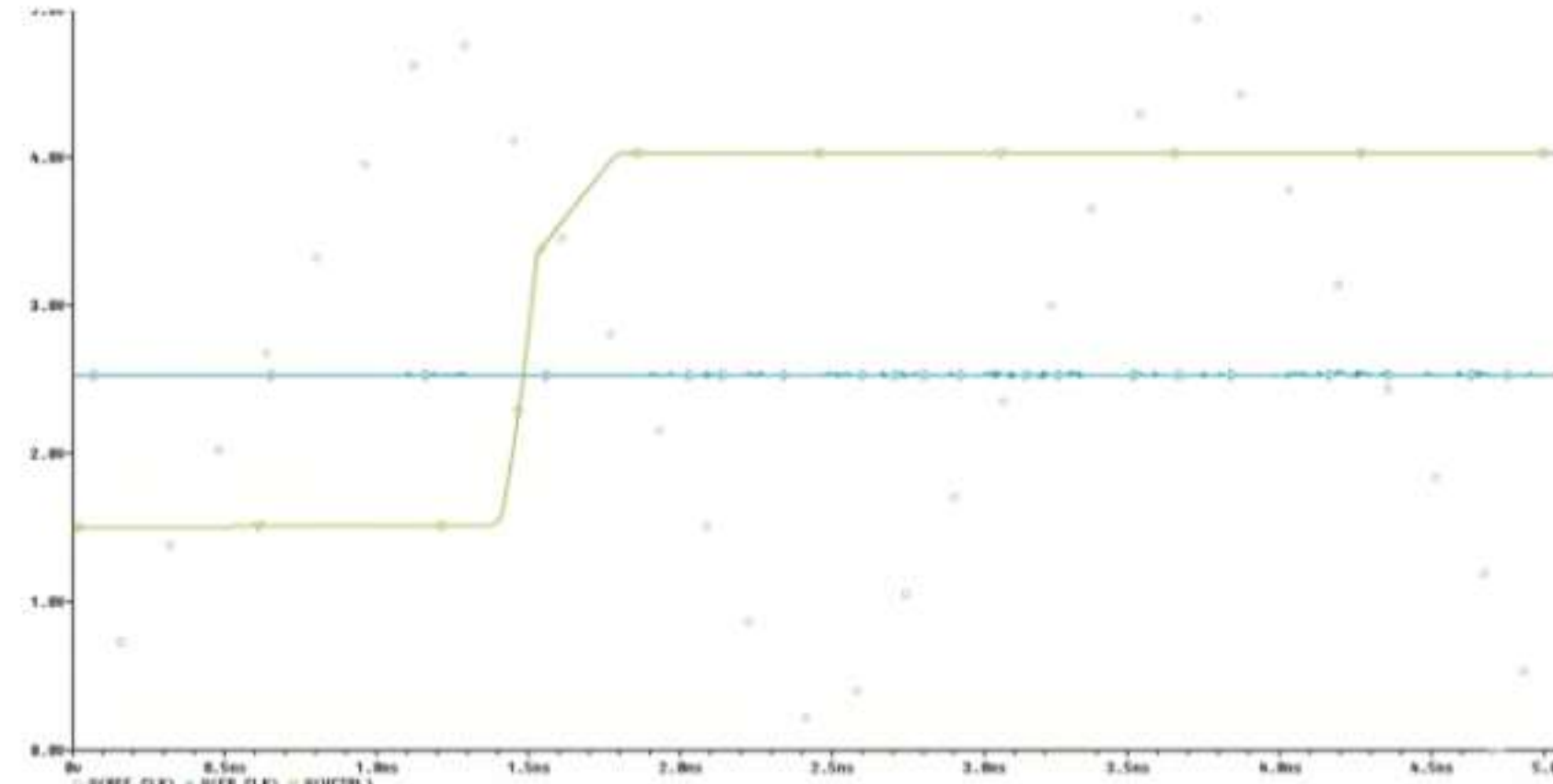


Figure 14: Loop filter integration. V_{ctrl} (blue trace) ramps up and stabilizes to a pure DC value, indicating successful frequency acquisition and phase lock.

System-Level Steady-State Lock

Figure 15 provides a macroscopic view of the PLL operating in steady-state lock. The V_{ctrl} line remains flat, proving the VCO frequency is perfectly stable. Furthermore, the ANALOG_VCC rail remains completely isolated from the aggressive digital switching of the VCO_OUT signal, validating the inclusion of the LM317 regulation block.

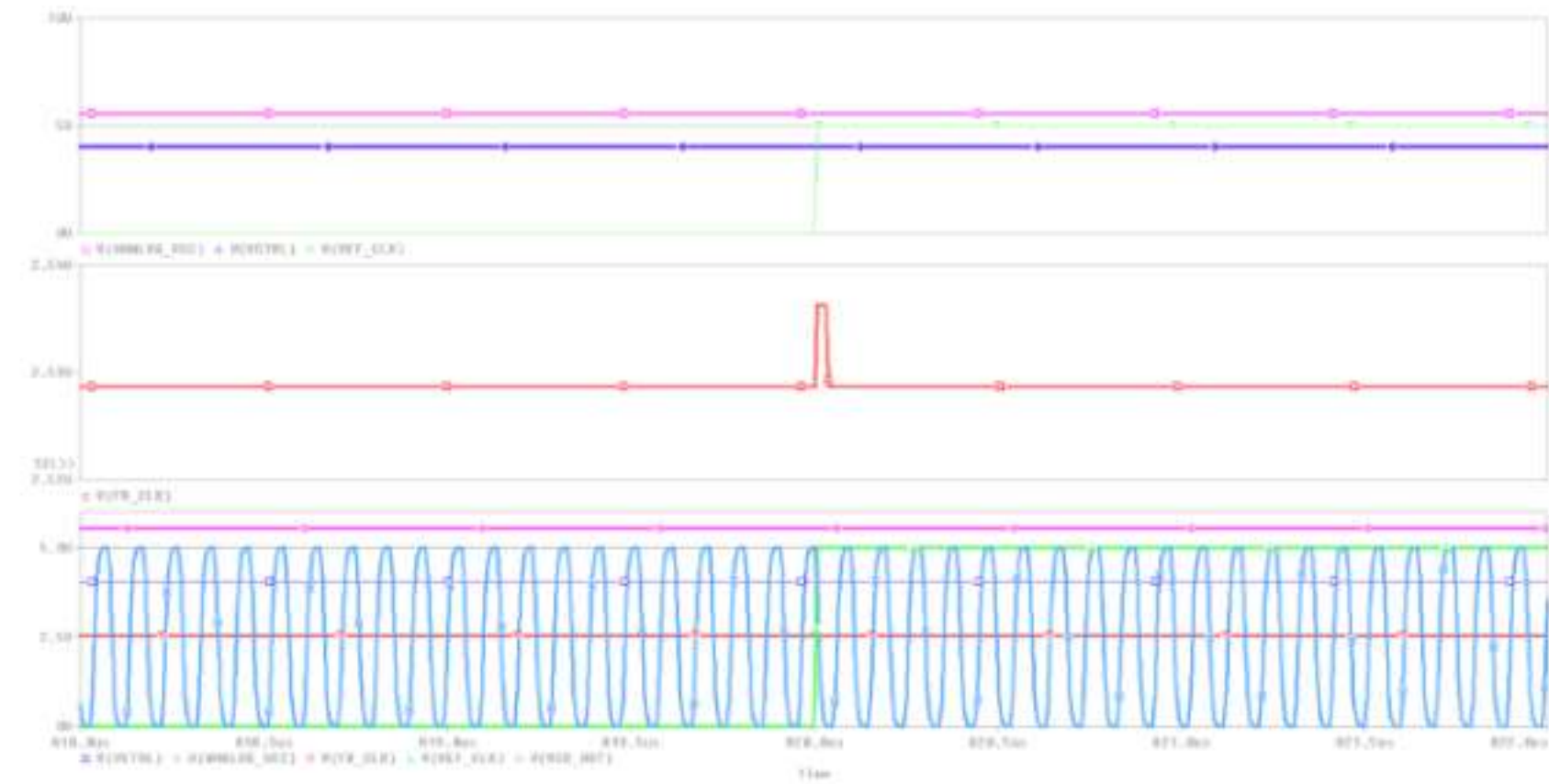


Figure 15: Complete system transient simulation. The high-speed VCO_OUT (light blue) runs stably, while the divided FB_CLK (red) perfectly tracks the incoming REF_CLK (green).

Experimental Setup and Hardware Instrumentation

To validate the theoretical models and Cadence simulations, the physical discrete PLL prototype was evaluated using standard laboratory benchtop instrumentation.

Stimulus and Power Parameters

The testbench required strict adherence to simulated voltage thresholds to ensure 74HC series logic operated correctly.

- **Power Delivery:** A TMI Systems DC Regulated Power Supply provided a stable 9.0 V master output. This rail

fed directly into the LM317 regulator block on the perf-board to be stepped down into the isolated analog supply.

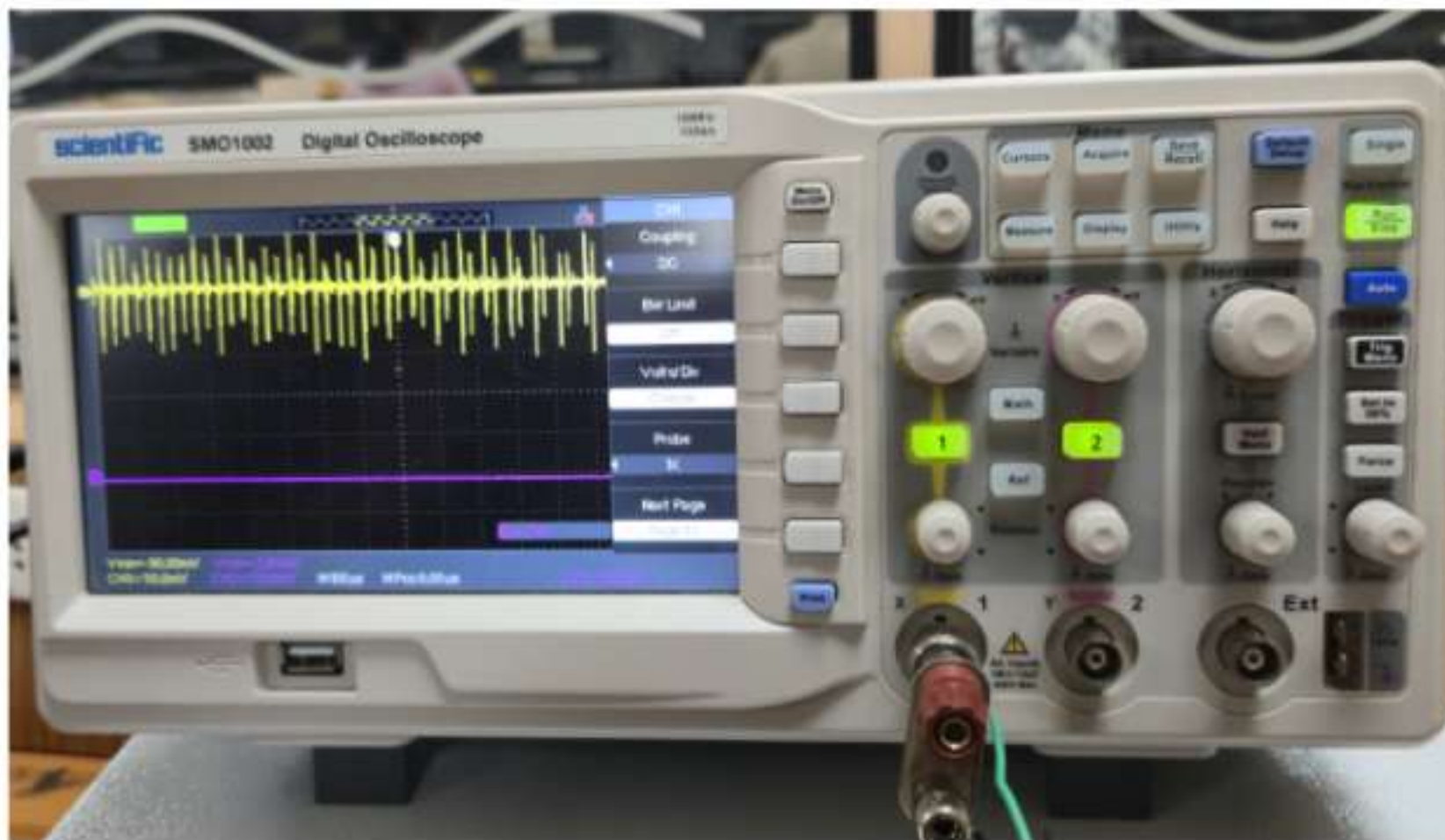
- **Reference Clock Generation:** A Keysight EDU33211A Waveform Generator synthesized the REF_CLK signal. Configured for a 50 kHz square wave with 5 V_{pp} amplitude and +2.5 V DC offset, ensuring clean 0 V–5 V swing without negative excursions.
- **Transient Capture:** A Scientific SMO1002 (100 MHz, 1 GSa/s) digital oscilloscope was employed for real-time node probing.



(a) TMI DC Power Supply



(b) Keysight Waveform Generator



(c) Scientific SMO1002 Oscilloscope

Figure 16: Laboratory instrumentation for physical circuit characterization. (a) Regulated supply provides 9 V input to LM317. (b) Waveform generator supplies 50 kHz CMOS reference clock ($V_{pp} = 5$ V, Offset = +2.5 V). (c) Oscilloscope captures transient dynamics and loop filter ripple.

Firmware Implementation

The complete Arduino/C++ source code is provided in Listing 1. It includes setup routines for the LEDC and PCNT peripherals, alongside the asynchronous web server serving the monitoring dashboard.

```
1 /*
2  * discrete_pll_esp32.ino
3  *
4  * ESP32 firmware for CSRO-VCO discrete PLL characterization.
5  *
6  * HARDWARE (from Notion project page verified pin mapping):
7  * GPIO18 LM339 Pin6 (REF_CLK, LEDC ch0, 50kHz, 50% duty, 3.3V)
8  * GPIO34 74HC04 Pin6 (VCO_OUT) via 1k+2k divider [input-only]
9  * GPIO35 TL072 Pin7 (Vctrl) via 1k+2k divider [input-only]
10 * GPIO21 OLED SDA (4.7k pull-up to 3.3V)
11 * GPIO22 OLED SCL (4.7k pull-up to 3.3V)
12 *
13 * LEVEL SHIFT (mandatory 5V IC outputs 3.3V ESP32):
14 * IC_OUT(5V) 1k ESP32_GPIO
15 * 2k
16 * GND
17 * Vgpio = 5V 2k/(1k+2k) = 3.33V
18 *
19 * Vctrl ADC scale:
20 * ADC reads divided voltage. True Vctrl = ADC_V (1 + 1k/2k) =
21 * ADC_V 1.5
22 * Max Vctrl = 4.5V ADC sees 3.0V (within ESP32 3.3V max )
23 *
24 * PCNT gate time = 100ms multiply count by 10 for Hz.
25 *
26 * WiFi: serves JSON at http://<ESP32-IP>/data
27 * serves live dashboard at http://<ESP32-IP>/
28 *
29 * PLL params (N=4, Fref=50kHz Fvco_target = 200kHz):
30 * Change N_DIVIDER and REF_FREQ_HZ to match jumper selection.
31 *
32 * Libraries required (Arduino IDE Library Manager):
33 * Adafruit SSD1306, Adafruit GFX
34 * driver/ledc.h, driver/pcnt.h ESP-IDF, bundled with esp32
35 * board package
36 */
37 #include <Arduino.h>
38 #include <Wire.h>
39 #include <WiFi.h>
40 #include <WebServer.h>
41 #include <Adafruit_SSD1306.h>
42 #include "driver/ledc.h"
43 #include "driver/pcnt.h"
44
45
46 // Pin definitions
47 #define REF_CLK_PIN 18 // LEDC output LM339 Pin6
48 #define VCO_IN_PIN 34 // PCNT input 74HC04 Pin6 (via divider)
49 #define VCTRL_PIN 35 // ADC1 ch7 TL072 Pin7 (via divider)
50 #define OLED_SDA 21
51 #define OLED_SCL 22
52
53 // PLL parameters
54 // Change N_DIVIDER to match 74HC163 jumper: QA=2, QB=4, QC=8,
55 // QD=16
56 #define N_DIVIDER 4
57 #define REF_FREQ_HZ 50000UL // 50kHz reference
58 #define FVCO_TARGET (N_DIVIDER * REF_FREQ_HZ) // 200kHz for N=4
59 #define LOCK_THRESHOLD 500 // 500Hz = locked
60 #define LOCK_HYSTERESIS 50 // consecutive samples for lock
61 // confirmation
62
63 // LEDC (reference clock)
64 #define LEDC_CHANNEL LEDC_CHANNEL_0
65 #define LEDC_TIMER LEDC_TIMER_0
66 #define LEDC_SPEED_MODE LEDC_HIGH_SPEED_MODE
67 #define LEDC_RESOLUTION LEDC_TIMER_8_BIT // 8-bit 256 steps,
68 // 50% = 128
69
70 // PCNT (VCO frequency measurement)
71 #define PCNT_UNIT PCNT_UNIT_0
72 #define PCNT_CHANNEL PCNT_CHANNEL_0
73 #define PCNT_GATE_MS 100 // count window in ms
74 // Fvco (Hz) = pcnt_count (1000 / PCNT_GATE_MS)
```



```

73 // ADC calibration
74 // ESP32 ADC nonlinearity warning: accurate 0.1V3.1V range.
75 // Vctrl via 1k+2k divider: max 3.0V at ADC within linear range
76
77 #define ADC_MAX_COUNTS 4095.0f
78 #define ADC_VREF 3.3f
79 #define VCTRL_DIVIDER_RATIO 1.5f // (1k+2k)/2k = 1.5 scale back
80 // up
81 // OLED
82 #define OLED_WIDTH 128
83 #define OLED_HEIGHT 64
84 #define OLED_ADDR 0x3C
85 Adafruit_SSD1306 display(OLED_WIDTH, OLED_HEIGHT, &Wire, -1);
86
87 // Web server
88 WebServer server(80);
89
90 // State
91 volatile float g_fvco = 0.0f;
92 volatile float g_vctrl = 0.0f;
93 volatile bool g_locked = false;
94 volatile int g_lock_count = 0;
95 volatile float g_kvco_est = 0.0f; // estimated Kvco from last
96 // two Vctrl/Fvco pairs
97 static float g_prev_vctrl = -1.0f;
98 static float g_prev_fvco = -1.0f;
99
100 // Function prototypes
101 void setupLEDC();
102 void setupPCNT();
103 float measureFvco();
104 float readVctrl();
105 void updateLockState(float fvco);
106 void updateOLED(float fvco, float vctrl, bool locked);
107 void handleRoot();
108 void handleData();
109 String buildHTML();
110
111 //
112 // setup()
113 void setup() {
114     Serial.begin(115200);
115     Serial.println("[PLL] Booting...");
116
117     // LEDC ref clock
118     setupLEDC();
119     Serial.printf("[PLL] REF_CLK: %lu Hz on GPIO%d\n",
120         REF_FREQ_HZ, REF_CLK_PIN);
121
122     // PCNT VCO counter
123     setupPCNT();
124     Serial.println("[PLL] PCNT ready on GPIO34");
125
126     // ADC
127     analogReadResolution(12);
128     analogSetAttenuation(ADC_11db); // full 0.3V range
129     Serial.println("[PLL] ADC: 12-bit, 11dB atten");
130
131     // OLED
132     Wire.begin(OLED_SDA, OLED_SCL);
133     if (!display.begin(SSD1306_SWITCHCAPVCC, OLED_ADDR)) {
134         Serial.println("[WARN] OLED not found continuing without
135             display");
136     } else {
137         display.clearDisplay();
138         display.setTextSize(1);
139         display.setTextColor(SSD1306_WHITE);
140         display.setCursor(0, 0);
141         display.println("PLL Monitor v1.0");
142         display.println("Initialising...");
143         display.display();
144         Serial.println("[PLL] OLED OK");
145     }
146
147     // WiFi
148     Serial.printf("[WiFi] Connecting to %s", WIFI_SSID);
149     WiFi.begin(WIFI_SSID, WIFI_PASSWORD);
150     int wifi_attempts = 0;
151     while (WiFi.status() != WL_CONNECTED && wifi_attempts < 20)
152     {
153         delay(500);
154         Serial.print(".");
155         wifi_attempts++;
156     }
157
158     if (WiFi.status() == WL_CONNECTED) {
159         Serial.printf("\n[WiFi] Connected: http://%s/\n", WiFi.
160             localIP().toString().c_str());
161         server.on("/", handleRoot);
162         server.on("/data", handleData);
163         server.begin();
164     } else {
165         Serial.println("\n[WiFi] Failed running offline (OLED +
166             Serial only)");
167     }
168
169     Serial.printf("[PLL] Target: Fvco = %lu Hz (N=%d, Fref=%lu
170         Hz)\n",
171         FVCO_TARGET, N_DIVIDER, REF_FREQ_HZ);
172     Serial.println("[PLL] Init complete.\n");
173 }
174
175 //
176 // loop()
177 //
178 void loop() {
179     // Measure
180     float fvco = measureFvco(); // Hz
181     float vctrl = readVctrl(); // true Vctrl in volts
182
183     // Lock detection
184     updateLockState(fvco);
185     bool locked = g_locked;
186
187     // Kvco estimation (running): F/V between last two valid
188     // samples
189     if (g_prev_vctrl > 0 && fabsf(vctrl - g_prev_vctrl) > 0.05f)
190     {
191         g_kvco_est = fabsf(fvco - g_prev_fvco) / fabsf(vctrl -
192             g_prev_vctrl);
193     }
194     g_prev_vctrl = vctrl;
195     g_prev_fvco = fvco;
196
197     // Update globals for web server
198     g_fvco = fvco;
199     g_vctrl = vctrl;
200
201     // Serial plotter (compatible with Arduino IDE Serial
202     // Plotter)
203     // Format: label:value shows all traces simultaneously
204     Serial.printf("Fvco:%.0f Ftarget:%lu Vctrl:%.3f Locked:%d
205         Kvco:%.0f\n",
206         fvco, FVCO_TARGET, vctrl, (int)locked,
207         g_kvco_est);
208
209     // OLED
210     updateOLED(fvco, vctrl, locked);
211
212     // Web server
213     if (WiFi.status() == WL_CONNECTED) {
214         server.handleClient();
215     }
216
217     // Gate time already consumed in measureFvco() no extra
218     // delay needed
219 }
220
221 //
222 // setupLEDC() 50kHz 50% duty on GPIO18 via ESP-IDF LEDC driver
223 //
224 void setupLEDC() {
225     ledc_timer_config_t timer_cfg = {
226         .speed_mode = LEDC_SPEED_MODE,
227         .duty_resolution = LEDC_RESOLUTION,
228         .timer_num = LEDC_TIMER,
229         .freq_hz = REF_FREQ_HZ,
230         .clk_cfg = LEDC_AUTO_CLK,
231     };
232     ledc_timer_config(&timer_cfg);
233
234     ledc_channel_config_t ch_cfg = {
235         .gpio_num = REF_CLK_PIN,
236         .speed_mode = LEDC_SPEED_MODE,
237         .channel = LEDC_CHANNEL,
238         .timer_sel = LEDC_TIMER,
239         .duty = 128, // 128/256 = 50% duty cycle
240         .hpoint = 0,
241     };
242     ledc_channel_config(&ch_cfg);
243 }

```



```

231 // Dynamically change reference frequency (e.g., for lock range
232 // sweep)
233 void setRefFreq(uint32_t freq_hz) {
234     ledc_set_freq(LEDC_SPEED_MODE, LEDC_TIMER, freq_hz);
235 }
236 //
237 // setupPCNT() hardware pulse counter on GPIO34 (input-only pin)
238 //
239 void setupPCNT() {
240     pcnt_config_t cfg = {
241         .pulse_gpio_num = VCO_IN_PIN,
242         .ctrl_gpio_num = PCNT_PIN_NOT_USED,
243         .lctrl_mode = PCNT_MODE_KEEP,
244         .hctrl_mode = PCNT_MODE_KEEP,
245         .pos_mode = PCNT_COUNT_INC, // count rising edges
246         .neg_mode = PCNT_COUNT_DIS,
247         .counter_h_lim = 30000,
248         .counter_l_lim = 0,
249         .unit = PCNT_UNIT,
250         .channel = PCNT_CHANNEL,
251     };
252     pcnt_unit_config(&cfg);
253     pcnt_set_filter_value(PCNT_UNIT, 100); // glitch filter:
254         // 400ns @ 240MHz
255     pcnt_filter_enable(PCNT_UNIT);
256     pcnt_counter_pause(PCNT_UNIT);
257     pcnt_counter_clear(PCNT_UNIT);
258     pcnt_counter_resume(PCNT_UNIT);
259 }
260 //
261 // measureFvco() gate-time frequency measurement using PCNT
262 // Blocks for PCNT_GATE_MS ms. Returns frequency in Hz.
263 //
264 float measureFvco() {
265     pcnt_counter_clear(PCNT_UNIT);
266     delay(PCNT_GATE_MS);
267     int16_t count = 0;
268     pcnt_get_counter_value(PCNT_UNIT, &count);
269     // count = pulses in PCNT_GATE_MS ms scale to Hz
270     return (float)count * (1000.0f / (float)PCNT_GATE_MS);
271 }
272 //
273 // readVctrl() 12-bit ADC on GPIO35, corrects for 1k+2k voltage
274 // divider
275 // Returns true Vctrl in volts.
276 //
277 float readVctrl() {
278     // Average 16 samples to reduce ADC noise
279     uint32_t sum = 0;
280     for (int i = 0; i < 16; i++) {
281         sum += analogRead(VCTRL_PIN);
282         delayMicroseconds(50);
283     }
284     float raw_avg = sum / 16.0f;
285     float adc_v = raw_avg * (ADC_VREF / ADC_MAX_COUNTS);
286     return adc_v * VCTRL_DIVIDER_RATIO; // scale back: 3.0V 4.5
287         // V
288 }
289 //
290 // updateLockState() hysteresis-based lock detection
291 // Requires LOCK_HYSTERESIS consecutive samples within
292 // LOCK_THRESHOLD Hz
293 //
294 void updateLockState(float fvco) {
295     float err = fabsf(fvco - (float)FVCO_TARGET);
296     if (err < LOCK_THRESHOLD) {
297         g_lock_count = min(g_lock_count + 1, LOCK_HYSTERESIS);
298     } else {
299         g_lock_count = max(g_lock_count - 2, 0); // faster unlock
300             // than lock
301     }
302     g_locked = (g_lock_count >= LOCK_HYSTERESIS);
303 }
304 //
305 // updateOLED() 4-line display
306 // Line 1: Fref / Ftarget
307 // Line 2: Fvco (measured)
308 // Line 3: Vctrl
309 // Line 4: LOCKED / HUNTING + Kvco estimate
310 //
311 void updateOLED(float fvco, float vctrl, bool locked) {
312     display.clearDisplay();
313     display.setTextSize(1);
314     display.setTextColor(SSD1306_WHITE);
315
316     display.setCursor(0, 0);
317     display.printf("Fref: %5lu Hz N=%d", REF_FREQ_HZ, N_DIVIDER);
318     ;
319
320     display.setCursor(0, 12);
321     display.printf("Fvco: %6.0f Hz", fvco);
322
323     display.setCursor(0, 24);
324     display.printf("Vctrl: %.3f V", vctrl);
325
326     display.setCursor(0, 36);
327     if (locked) {
328         display.print("Status: LOCKED ");
329     } else {
330         float err = fvco - (float)FVCO_TARGET;
331         display.printf("HUNTING %+.0fHz", err);
332     }
333
334     display.setCursor(0, 48);
335     if (g_kvco_est > 0) {
336         display.printf("Kvco: %.0f kHz/V", g_kvco_est / 1000.0f);
337     } else {
338         display.print("Kvco: measuring...");
339     }
340
341     display.display();
342 }

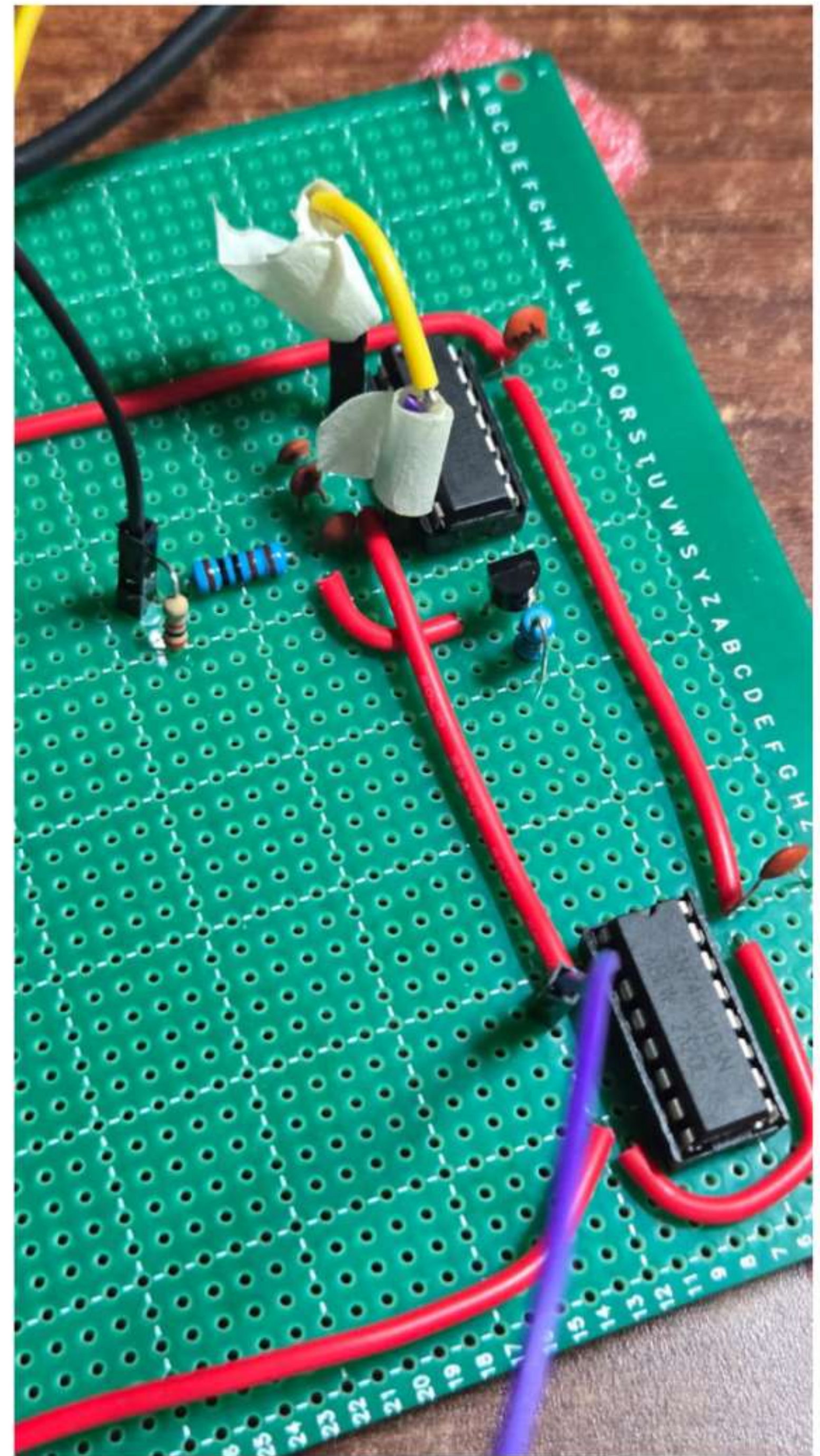
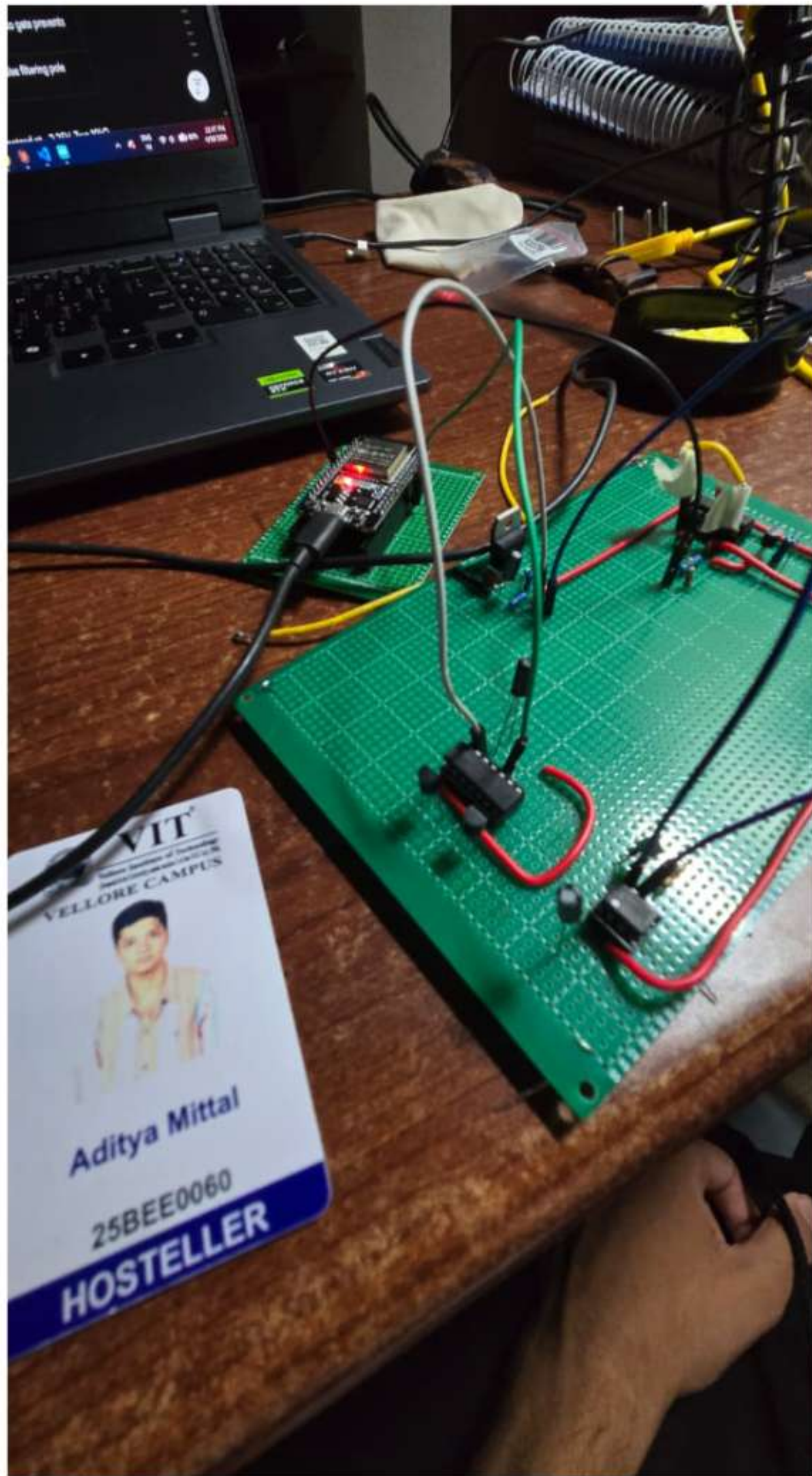
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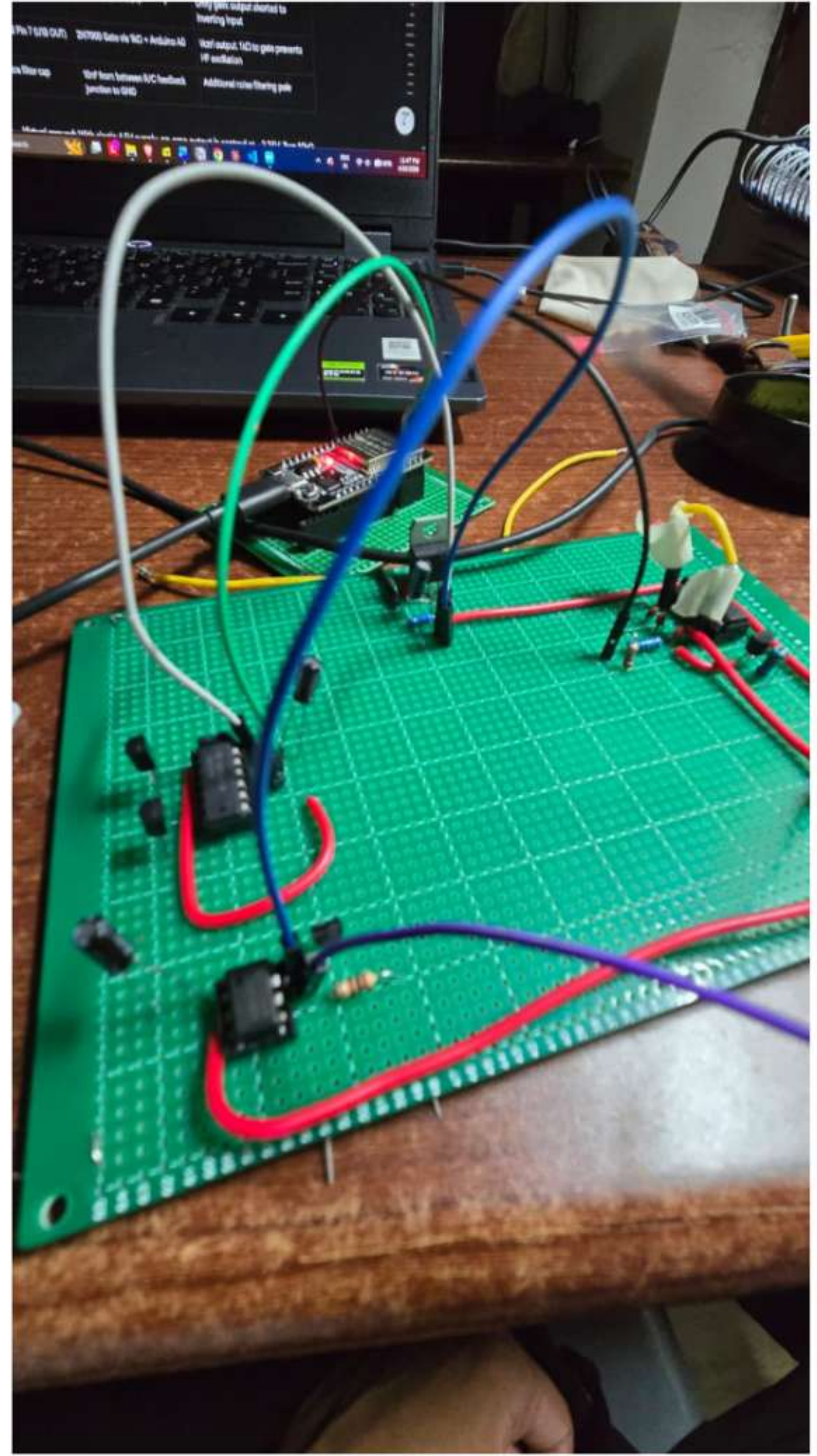
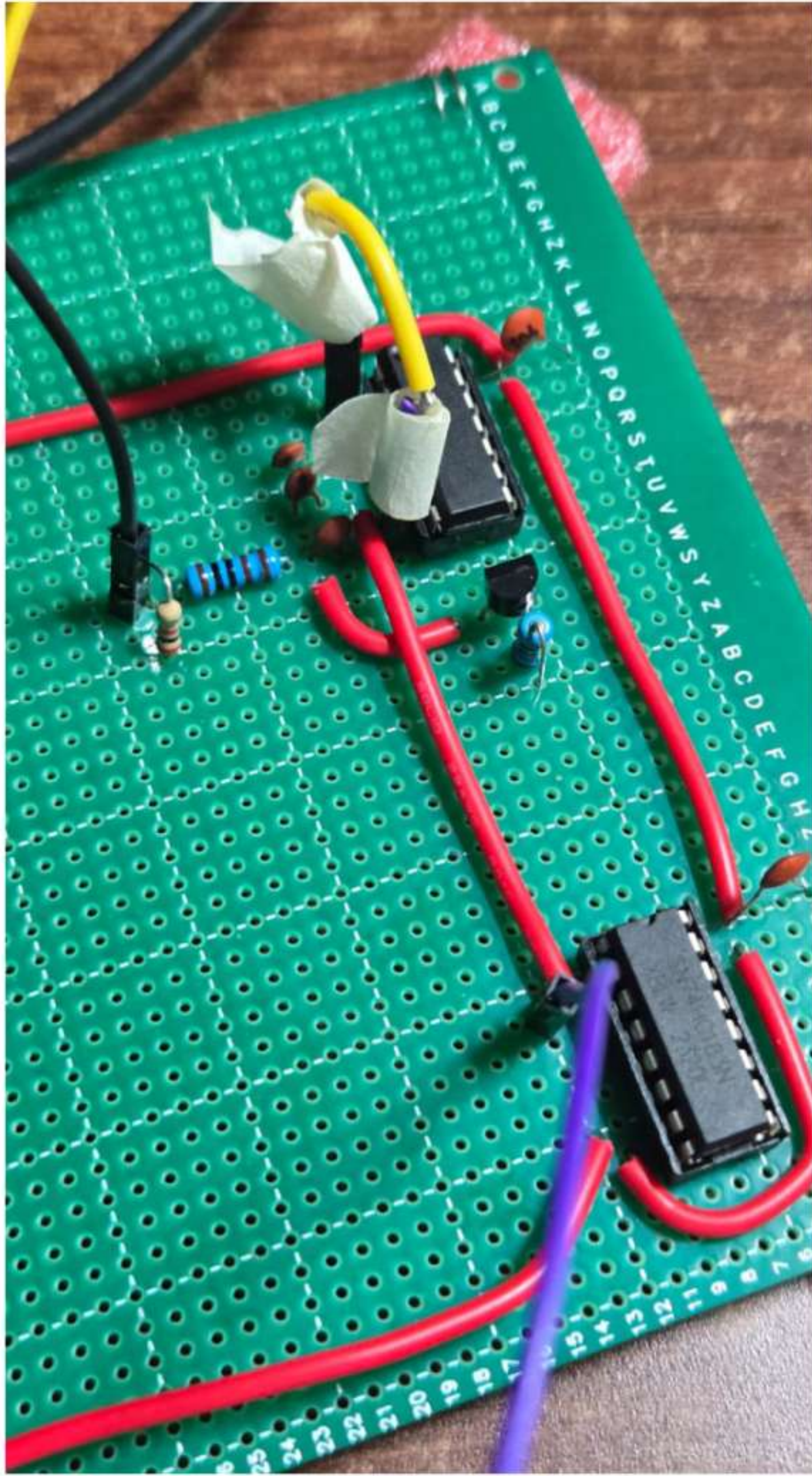
Listing 1: ESP32 PLL Characterization Firmware

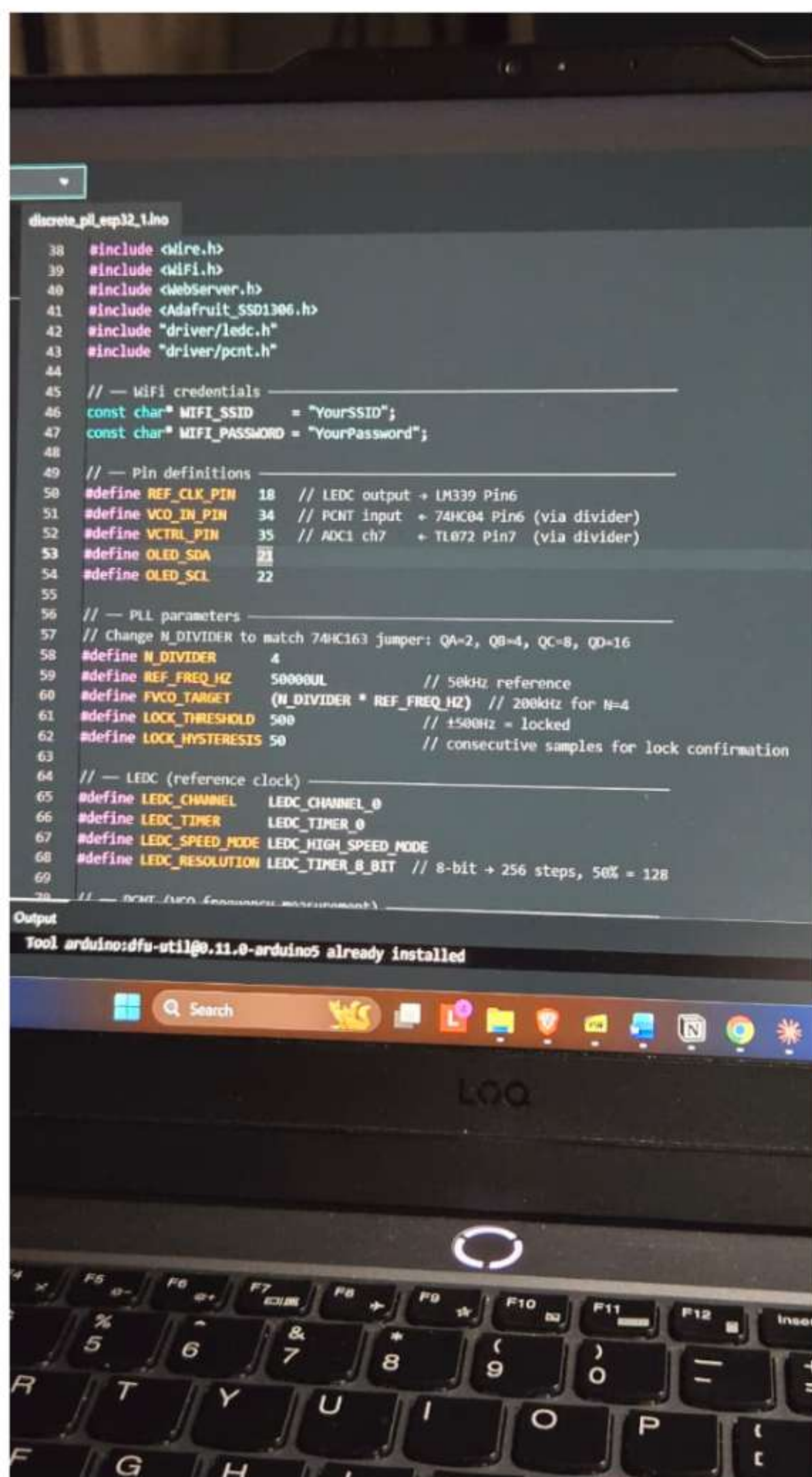
Hardware Implementation and Real-World Results

Following the validation of the Cadence PSpice simulations, the discrete architecture was successfully synthesized onto a 12×18 cm FR4 perfboard.

The physical hardware demonstrated successful phase acquisition and frequency locking at the target 50 kHz reference frequency. However, compared to the idealized transient simulations, the physical control voltage (V_{ctrl}) exhibited higher steady-state ripple, and the VCO output displayed measurable phase jitter. These deviations are expected in macroscopic discrete designs and are primarily attributed to component tolerances, BJT mismatch in the charge pump, and the inherent physical limitations of the perfboard medium.







Parasitic and Antenna Effects in Discrete Design

While integrated circuits (ICs) operate within heavily shielded, sub-micron silicon environments, this macroscopic prototype is highly susceptible to electromagnetic interference (EMI) and parasitic elements.

Stray Capacitance and Lead Inductance

The standard 2.54 mm pitch perfboard introduces significant parasitic routing elements. Every copper pad and exposed jumper wire acts as a tiny capacitor ($\approx 1\text{--}2\text{ pF}$ per node) to adjacent traces. Furthermore, the physical lengths of the through-hole component leads introduce parasitic series inductance. At high-speed digital switching nodes (such as the 74HC04 inverter outputs running at MHz frequencies), this inductance (L) and capacitance (C) create unintentional LC tank circuits, causing the ringing and overshoot observed on the oscilloscope traces.

The Perfboard as an Antenna

Unlike a multi-layer Printed Circuit Board (PCB) with a dedicated, unbroken ground plane, the point-to-point soldering on a perfboard creates large ground loops. These loops act as receiving antennas for ambient RF noise (such as 50 Hz mains hum and Wi-Fi signals). Additionally, the high slew-rate edges of the discrete digital divider inject switching noise directly into the substrate. Without a dedicated ground plane to absorb this energy, the noise couples into the highly sensitive analog V_{ctrl} node, directly manifesting as phase jitter at the VCO output. This physical phenomenon perfectly illustrates the critical importance of isolated analog and digital power domains in modern mixed-signal SoC layouts.

Conclusion

This project successfully bridged the gap between theoretical microelectronics and tangible hardware by physically realizing a Discrete Mixed-Signal Phase-Locked Loop. By replicating sub-micron CMOS topologies—specifically the Current-Starved Ring Oscillator and the BJT-based Charge Pump—at a macroscopic scale, the intricate dynamics of phase-frequency detection, active PI loop filtering, and control voltage integration were directly observed and characterized.

The system successfully achieved steady-state lock, demonstrating a VCO tuning gain of $K_{VCO} \approx 380\text{ kHz/V}$. Furthermore, the integration of the ESP32 microcontroller's hardware PCNT and LEDC peripherals provided a robust, modern telemetry interface for real-time system analysis. Ultimately, this discrete prototype serves as a highly effective educational platform, visualizing the complex mixed-signal phenomena that typically remain abstracted within highly integrated silicon.

Project Repository and Raw Data

To ensure complete transparency and reproducibility, all raw project data has been made open-source. The high-resolution Cadence OrCAD schematics, LTspice/PSpice simulation profiles, ESP32 C++ telemetry firmware, and physical hardware demonstration videos are available in the project repository.

Project Link: <https://drive.google.com/drive/u/2/folders/1eFnXQYgeTBe0HpKQcLploGk9TzpmBL2f>

Formatting Acknowledgment

The structural formatting and typesetting of this technical report were executed utilizing the LaTeX document preparation system via the Overleaf platform, specifically adapting the standard IEEE Transactions double-column template. Artificial Intelligence was utilized strictly in a minimal capacity to assist with the LaTeX code generation for complex

image alignments, matrix grids, and margin constraint resolutions. All circuit design, mathematical analysis, schematic capture, physical soldering, and core text generation were performed entirely independently.

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